



Chapter 16 Oscillations

The world is full of simple harmonic oscillators. Once we understand completely the SHO, we can explore the world of problems it solves.

The word harmonic refers to music: musical instruments generally vibrate harmonically.



Key words and Terms

harmonic motion

谐运动

simple harmonic motion (SHM)

简谐运动

simple harmonic oscillator (SHO)

简谐振子

restoring force

回复力

equilibrium position

平衡位置

amplitude

振幅

angular frequency

角频率

natural frequency

固有频率

phase

相位

phase constant (phase angle, initial phase)

相位常数

rotation vector

旋转矢量

forced oscillation

受迫振动

resonance

共振

resonant frequency

共振频率

initial conditions

初始条件

damped harmonic motion

阻尼谐振动

critical damping

临界阻尼

simple pendulum

单摆

physical pendulum

复摆

Chapter 16 Oscillations

16.1 Simple harmonic motion

16.2 Describing for SHM

16.3 Energy in SHM

16.4 Pendulums

16. 5 Composition of SHM

***16.6 Damped Harmonic Motion**

***16.7 Forced Oscillations and Resonance**

振动：任何物理量（位移、密度、压强、电流、电压、场强等）的值在某一定值附近作往复变化，都可以称之为振动。

机械振动：物体在其平衡位置附近往复运动。



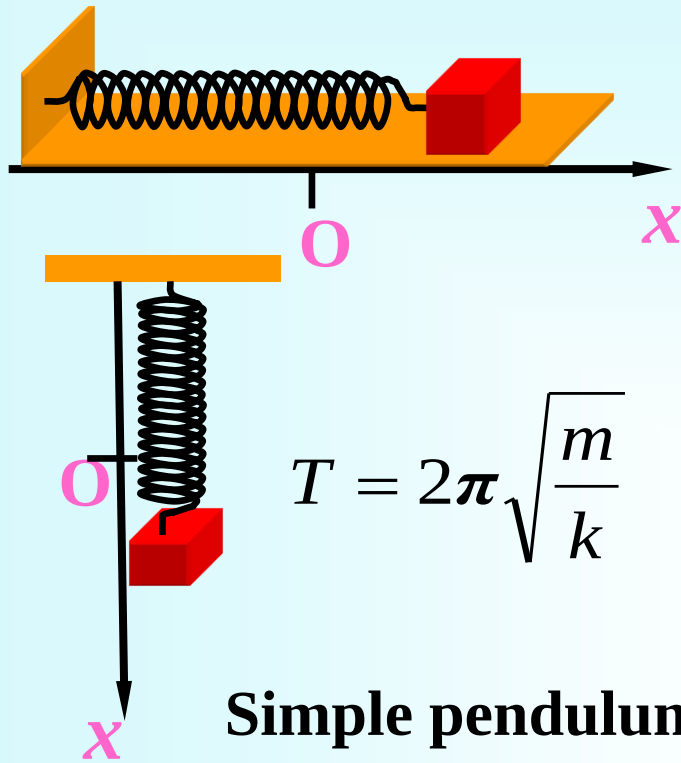
{ 受迫振动
自由振动

{ 阻尼自由振动
无阻尼自由振动

无阻尼自由谐振动
(简谐振动 SHM)

无阻尼自由非谐振动

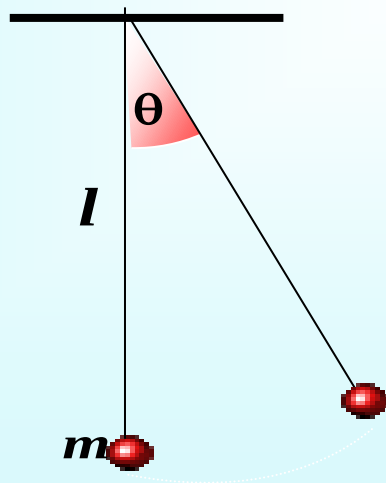
Simple harmonic oscillator (SHO)



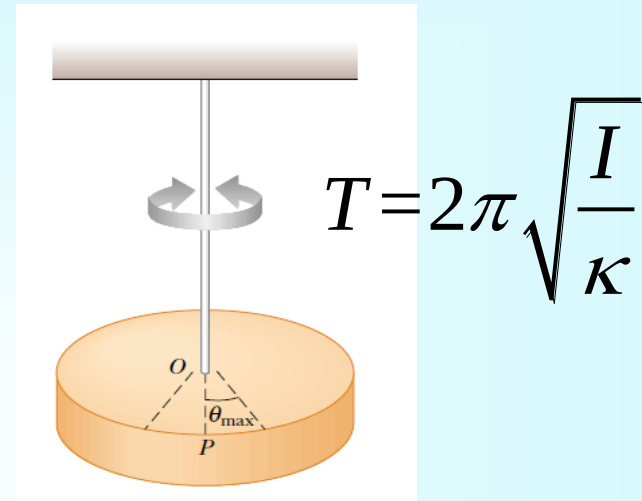
$$T = 2\pi \sqrt{\frac{m}{k}}$$

Simple pendulum

$$T = 2\pi \sqrt{\frac{L}{g}}$$

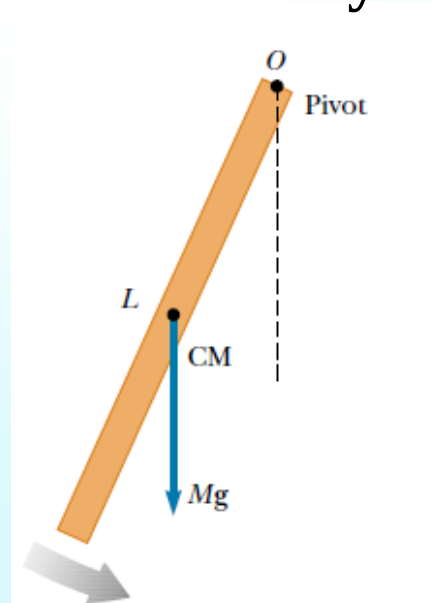


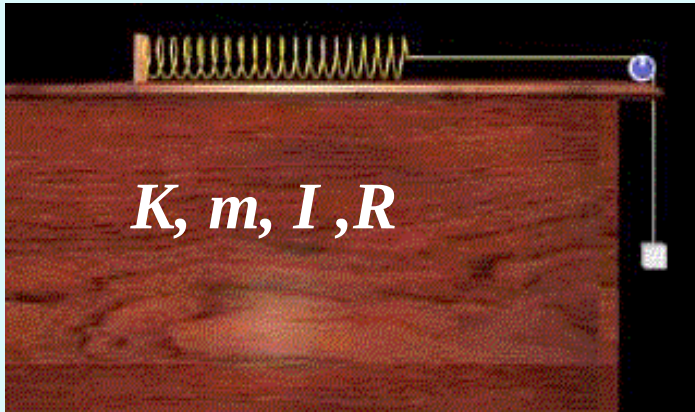
Torsion pendulum



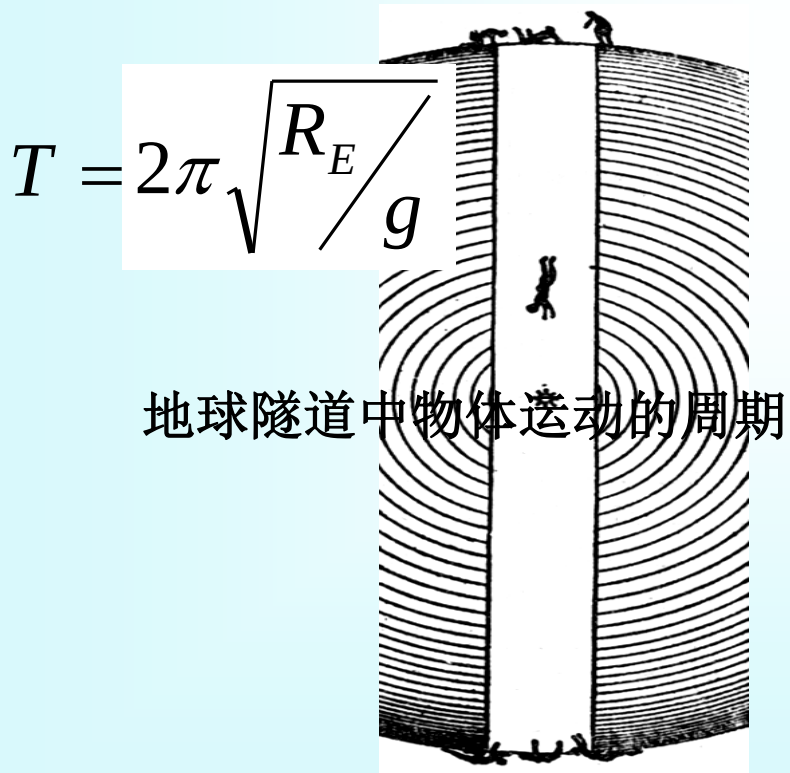
Physical pendulum

$$T = 2\pi \sqrt{\frac{I}{mgh}}$$

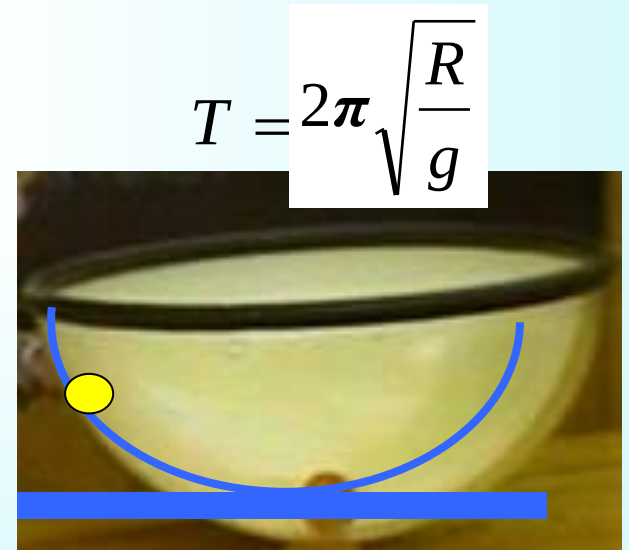




$$T = 2\pi \sqrt{\frac{m + I / R^2}{k}}$$



$$T = 2\pi \sqrt{\frac{R_E}{g}}$$



§ 16.1 Simple Harmonic Motion (SHM)



1. **Simple** Harmonic Motion (SHM):

An motion in which the **net** force is directly proportional to the negative of the displacement. Such system is called simple harmonic oscillator (SHO).

$$F = -k_{\text{effective}}x$$

Characteristics

(1) Equal amplitude

(2) Periodic $x(t)=x(t+T)$



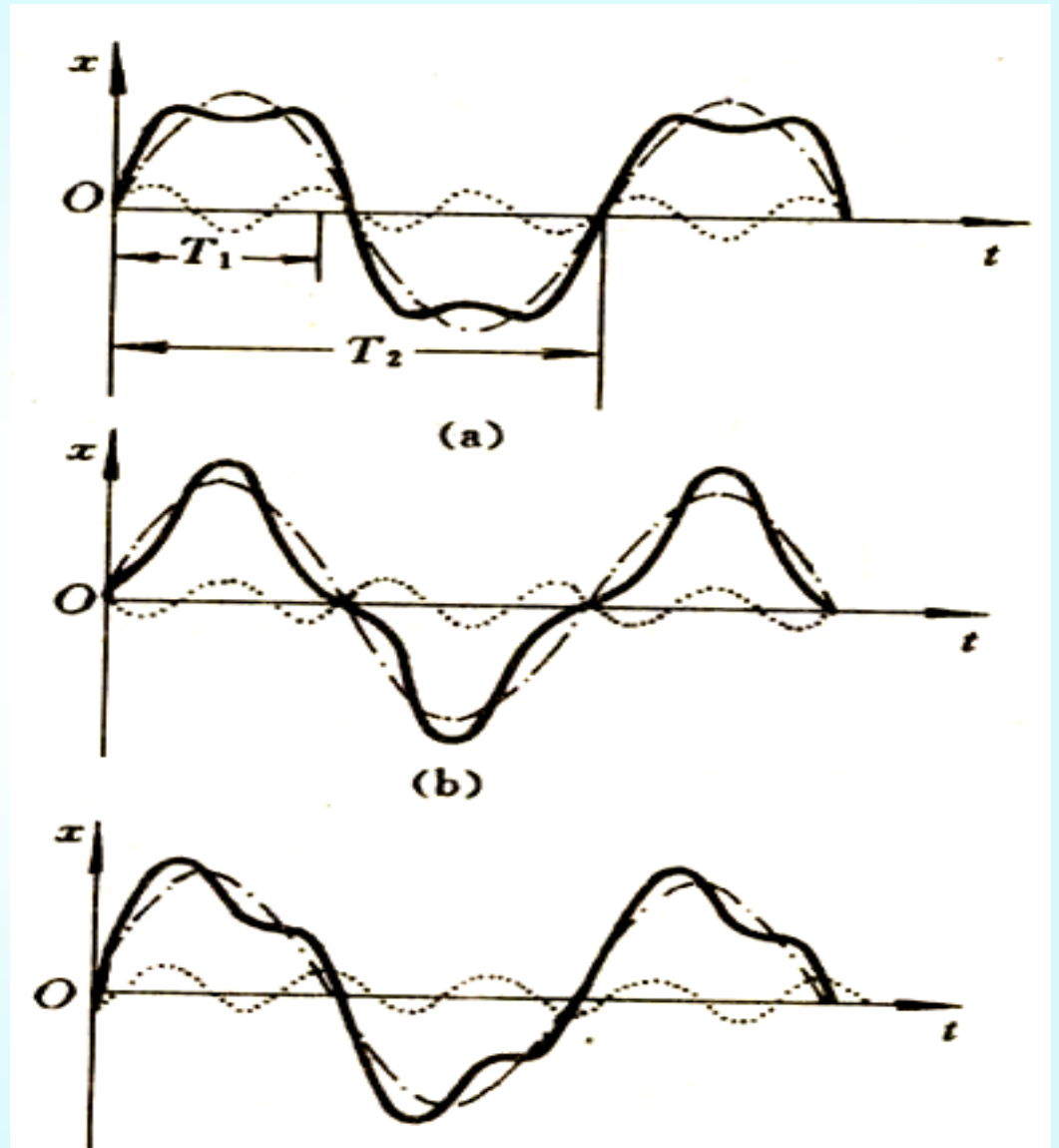
$$x = A \cos(\omega t + \varphi')$$

$$x = A \sin(\omega t + \varphi)$$



We see the importance of considering SHM: because any other wave shape can be considered a sum of such **pure sinusoidal wave**.

复杂振动可视为
几个简谐振动的合成



16.2 Describing for *SHM*

1. Motion equation

$$\left. \begin{array}{l} F = -k_{\text{effective}} x \\ F = m \frac{d^2 x}{dt^2} \end{array} \right\} \frac{d^2 x}{dt^2} + \frac{k}{m} x = 0 \quad \omega^2 \equiv \frac{k}{m} \quad \longrightarrow \quad \frac{d^2 x}{dt^2} + \omega^2 x = 0$$

Motion equation

$$a = -\omega^2 x$$

The solution of the motion equation of SHM can be

$$x(t) = A \cos(\omega t + \varphi)$$

The simple harmonic motion is the motion executed by a particle subject to a force that is proportional to the displacement of the particle but **opposite** in sign.

Question

拍皮球时，皮球的运动是简谐振动吗？
(设皮球与地面的碰撞是弹性碰撞)



2. A, ω, φ

Displacement at time t Phase 相位

$$x(t) = A \cos(0 + \varphi)$$

Amplitude phase constant or phase angle $[-\pi, \pi]$

(a) Amplitude (振幅): $A(x_m)$, unit: Hz

The **greatest distance** from the equilibrium point.

(b) Phase (相位): $\omega t + \varphi$ (function of time t), unit: rad

(c) (Natural) frequency (频率) and period(周期): f , unit: Hz; T , unit: s

The number of complete cycles per second.

(d) Angular frequency(圆频率): ω , unit: Hz

$$\omega = 2\pi f = \frac{2\pi}{T}$$

$T(f, \omega)$ is determined by the system, while A and φ are dependent of initial conditions.

3.Displacement, velocity and acceleration of SHM

$$x(t) = A \cos(\omega t + \varphi)$$

$$v = \frac{dx}{dt} = -A\omega \sin(\omega t + \varphi)$$

$$a = \frac{d^2x}{dt^2} = -A\omega^2 \cos(\omega t + \varphi)$$

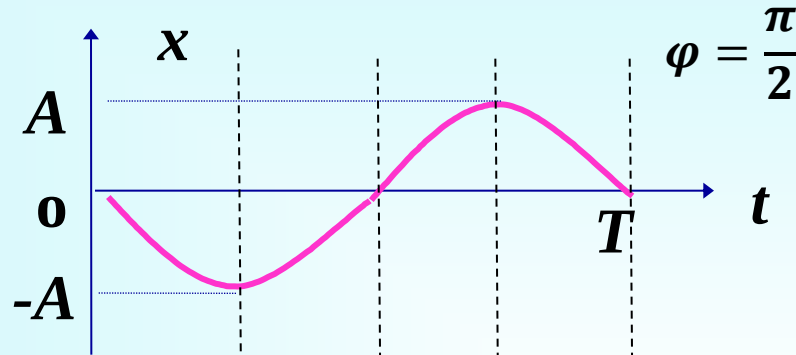
$$F = -kA \cos(\omega t + \varphi)$$

$x(t)$, $v(t)$, $a(t)$
and also $F(t)$
are all periodic
function of
time

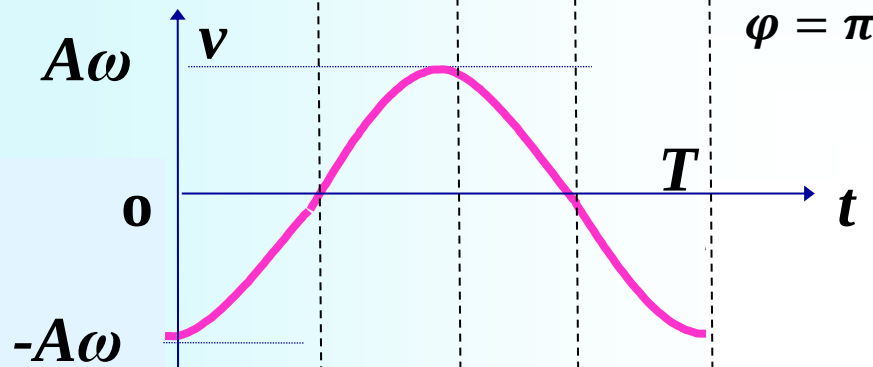
$$\omega^2 \equiv k / m$$

Once we have $x(t)$ for a particular system, we can find everything there is to know about the motion of that system.

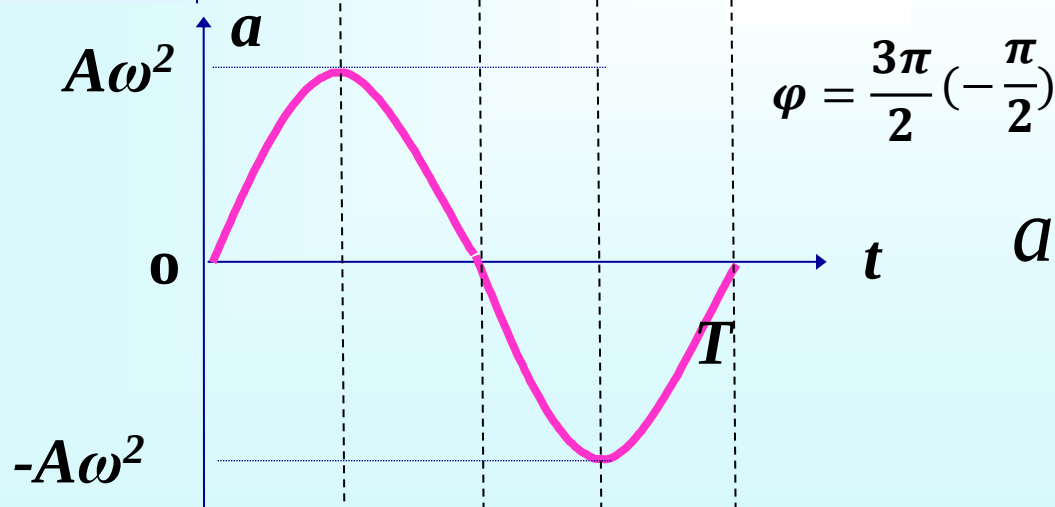
Displacement, velocity, acceleration function



$$x(t) = A \cos(\omega t + \varphi)$$

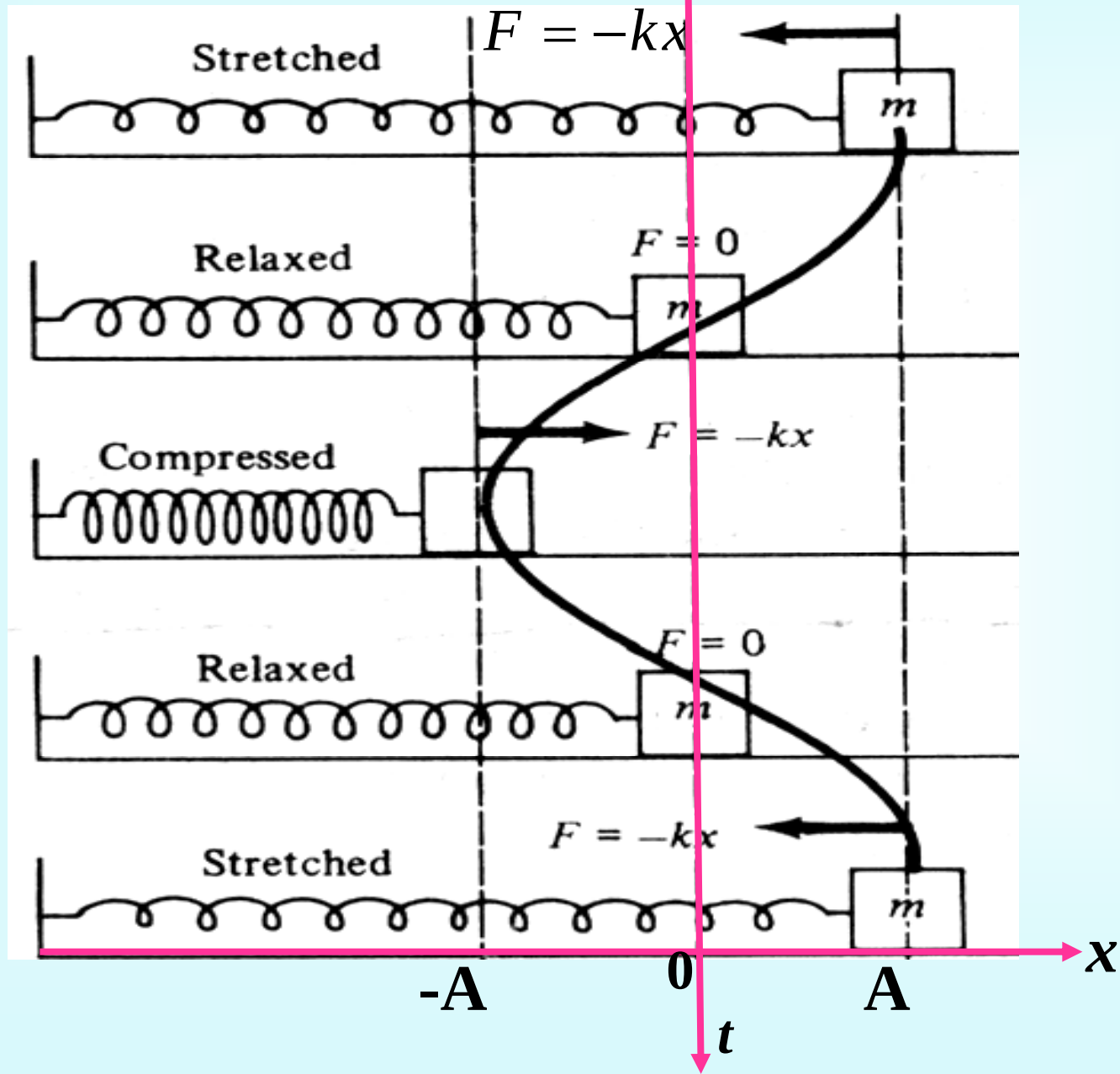


$$v = -v_m \sin(\omega t + \varphi)$$

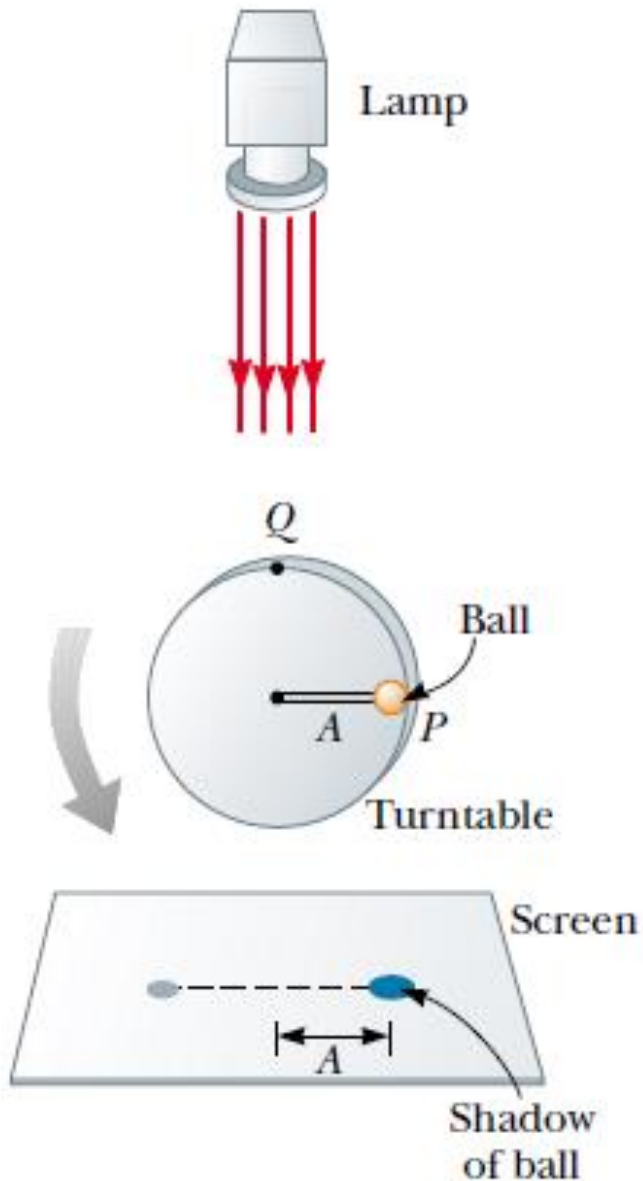


$$a = -a_m \cos(\omega t + \varphi)$$

Snapshots of a mass oscillating on the end of a spring



4. Rotation vector in plane

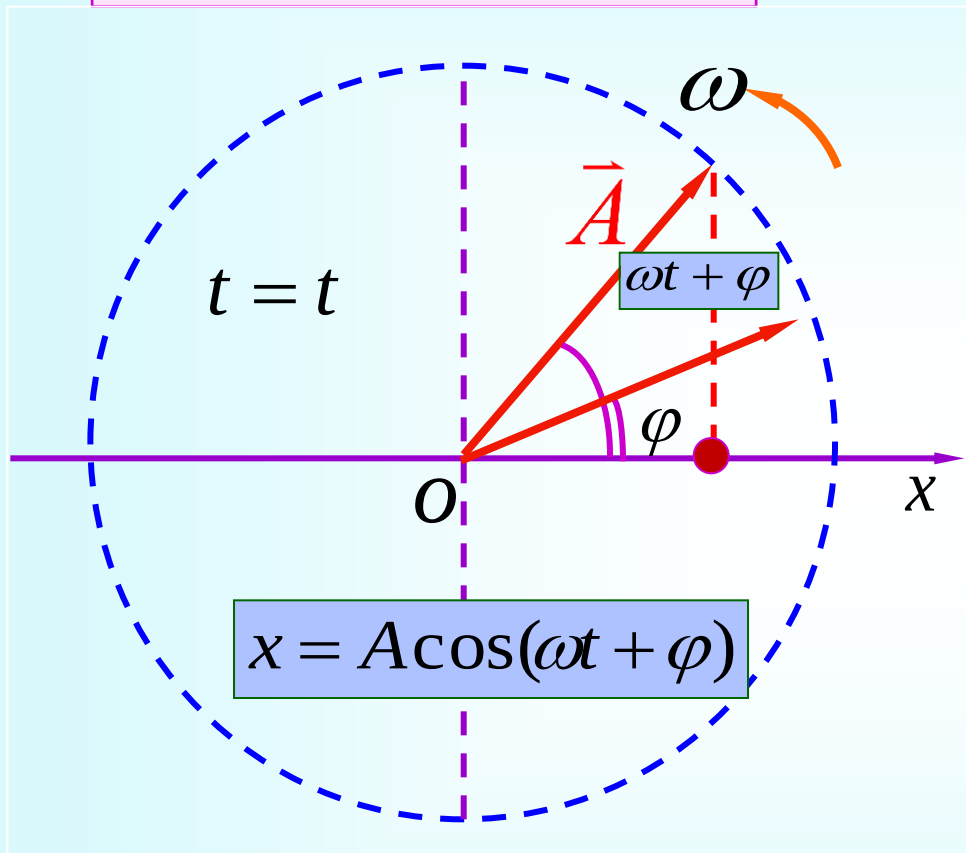


An experimental setup for demonstrating the connection between **simple harmonic motion** and **uniform circular motion**.

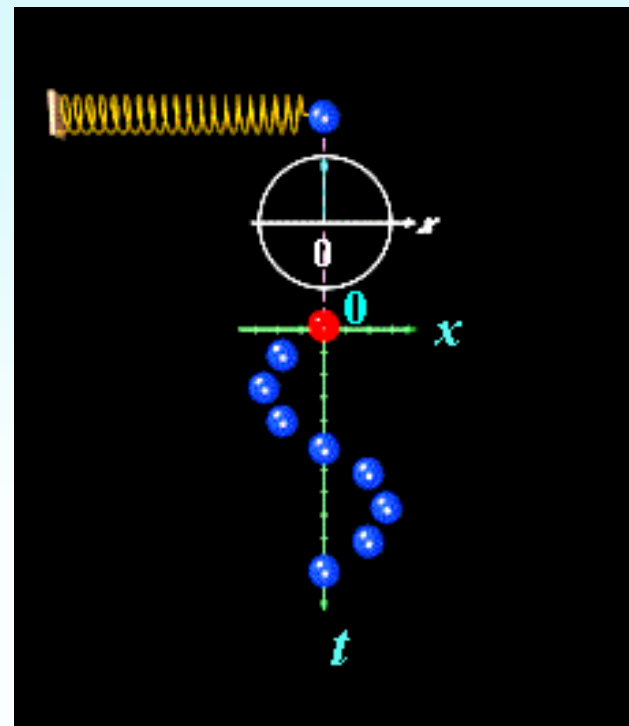
As the ball rotates on the turntable with constant angular speed, its shadow on the screen moves back and forth in simple **harmonic motion**.

Rotation vector

$$x = A\cos(\omega t + \varphi)$$

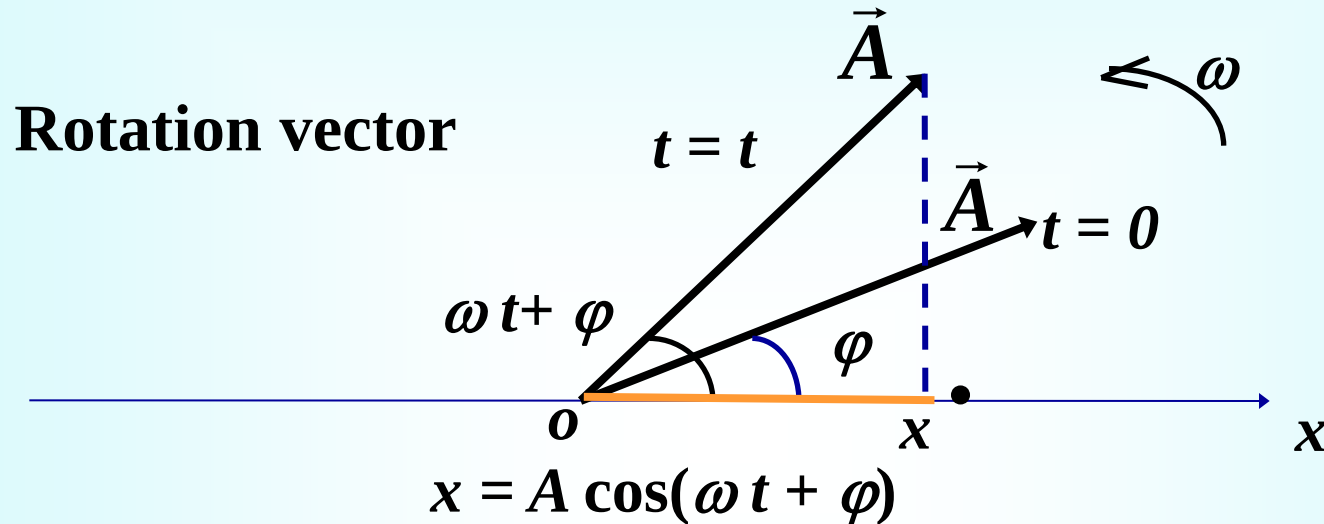


以 O 为原点旋转矢量 $\vec{A}$ 的端点在 x 轴上的投影点的振动为简谐振动。



矢量 $\vec{A}$ 其模等于振幅 A ，**逆时针**方向的匀角速转动，其角速度 ω ， $\vec{A}$ 这个矢量就叫做**旋转矢量**。

The oscillating motion is
one **component** of circular motion.



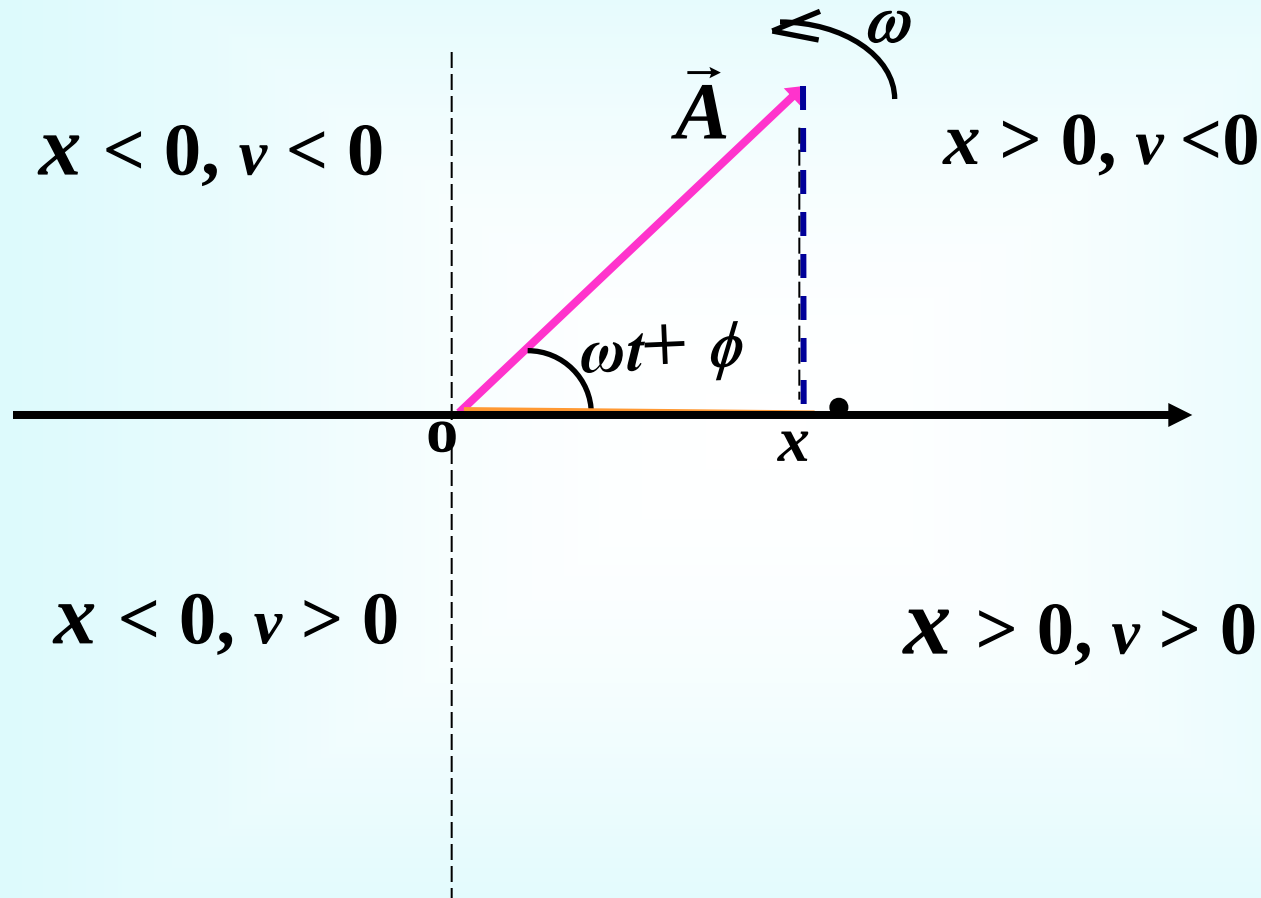
φ – Phase angle or phase constant (phase at $t=0$)

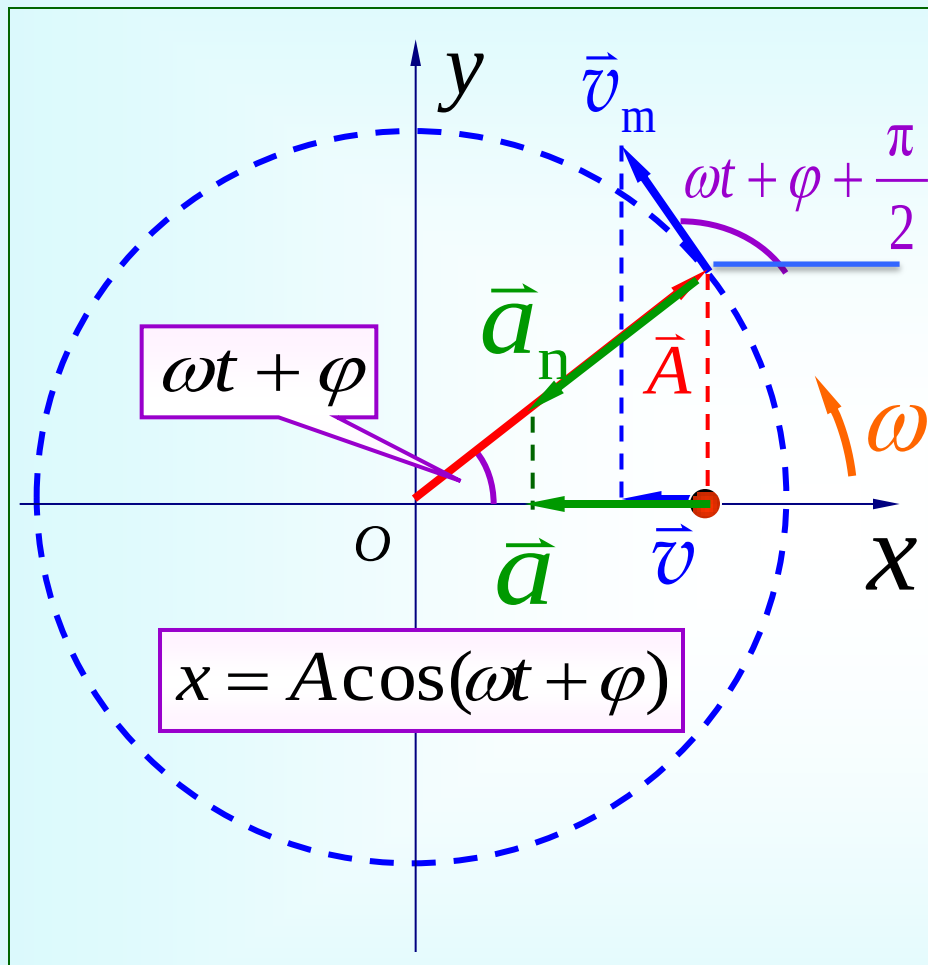
$\omega t + \varphi$ – Phase of time t

Rotation vector does **not** undergo the SHM.

它矢端在x轴上的投影点在x轴上作谐振动。

$$x = A \cos (\omega t + \phi)$$





$$v_m = A\omega$$

$$v = A\omega \cos\left(\omega t + \varphi + \frac{\pi}{2}\right)$$

$$v = -A\omega \sin(\omega t + \varphi)$$

$$a_n = A\omega^2$$

$$a = A\omega^2 \cos(\omega t + \varphi + \pi)$$

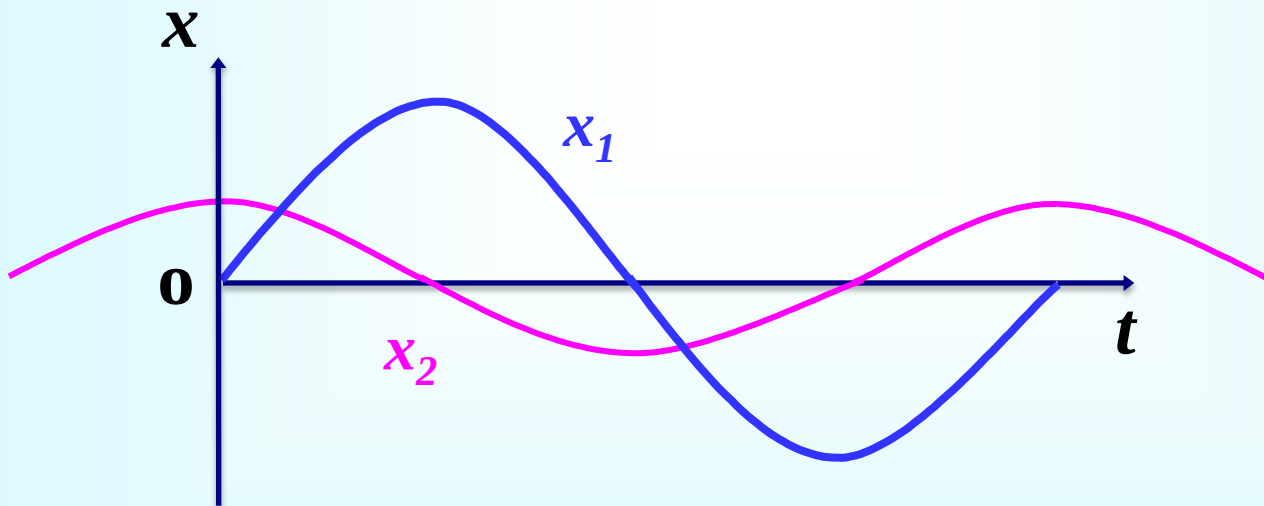
$$a = -A\omega^2 \cos(\omega t + \varphi)$$

矢端在 t 时刻的速度及加速度向 x 轴投影就是该时刻振动质点的速度和加速度。

5. 超前或落后

若 $\Delta\varphi = \varphi_2 - \varphi_1 > 0$, 则 x_2 比 x_1 较早达到正最大, 称 x_2 比 x_1 超前 (或 x_1 比 x_2 落后)。

超前和落后以 $|\Delta\varphi| < \pi$ 的相位角来判断



思考： 简谐振动的位移、速度、加速度谁超前？

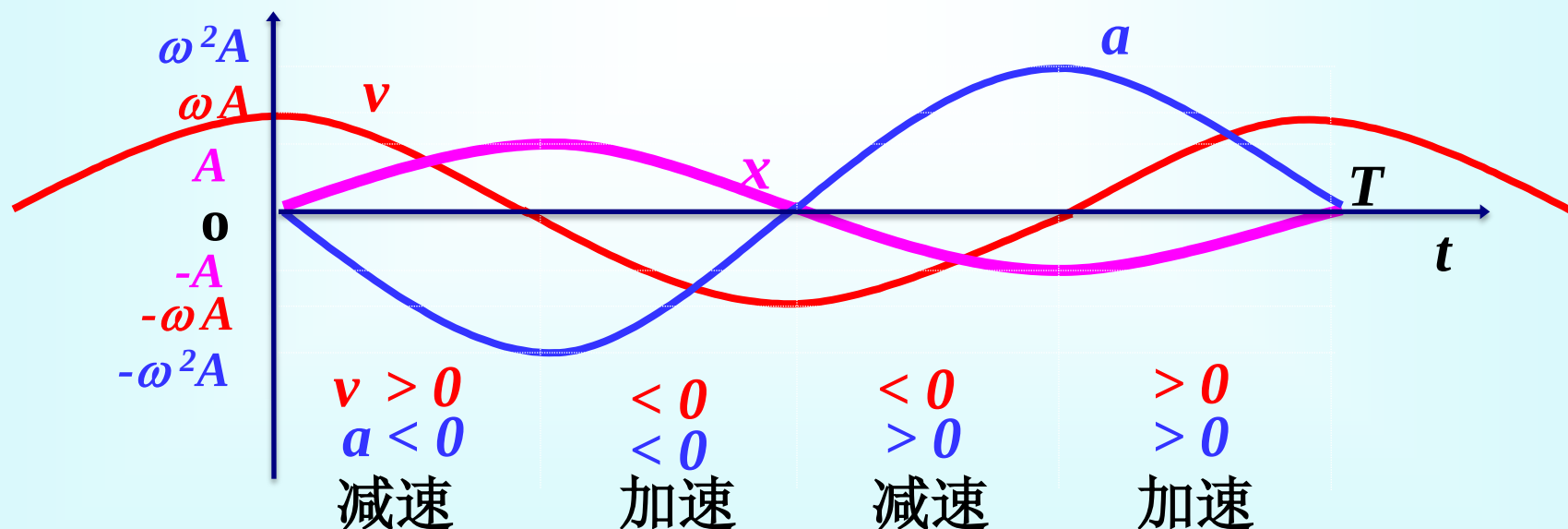
位移 $x(t) = A \cos(\omega t + \varphi)$

速度 $v = -\omega A \sin(\omega t + \varphi) = \omega A \cos(\omega t + \varphi + \frac{\pi}{2})$

加速度 $a = \omega^2 A \cos(\omega t + \varphi + \pi)$

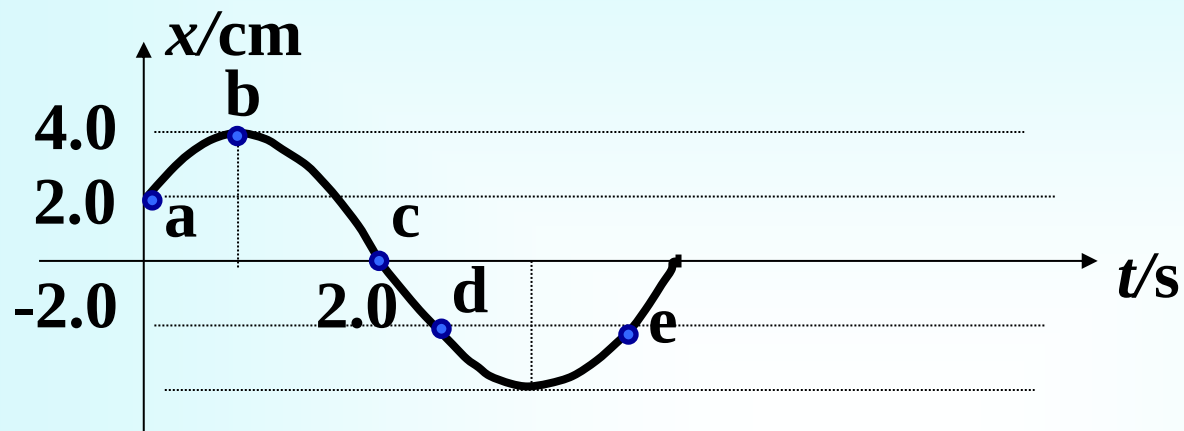
v 比 x 领先 $\pi/2$

a 比 v 领先 $\pi/2$





Determine: **Phase** of a, b, c, d, e,



$$a : -\frac{1}{3}\pi$$

$$b : 0$$

$$c : \frac{1}{2}\pi$$

$$d : \frac{2}{3}\pi$$

$$e : -\frac{2}{3}\pi$$

Example 1 The position of a SHO as a function of time is given by $x=3.8\cos(7\pi t/4+\pi/6)\text{m}$ where t is in seconds and x in meters. Find (a) the **period** and **frequency**, (b) the **position** and **velocity** at $t=0$, and (c) the **velocity** and **acceleration** at $t=2.0\text{s}$.

Solution:

$$(a) \quad T = \frac{2\pi}{\omega} = 2\pi / \frac{7}{4}\pi = \frac{8}{7} \text{ s} \quad f = \frac{1}{T} = 0.875 \text{ Hz}$$

$$t = 0 \quad x = 3.8 \cos \frac{\pi}{6} = 3.29 \text{ m}$$

$$v = \frac{dx}{dt} = -A\omega \sin(\omega t + \varphi) = -3.8 \times \left(\frac{7}{4}\pi\right) \sin \frac{\pi}{6} = -10.4 \text{ m/s}$$

$$(b) \quad t = 2 \text{ s} \quad v = -3.8 \times \left(\frac{7}{4}\pi\right) \sin\left(\frac{7}{4}\pi \times 2 + \frac{\pi}{6}\right) = 18 \text{ m/s}$$

$$a = \frac{dv}{dt} = -A\omega^2 \cos(\omega t + \varphi) = -3.8 \times \left(\frac{7}{4}\pi\right)^2 \cos\left(\frac{7}{4}\pi \times 2 + \frac{\pi}{6}\right) = -56.8 \text{ m/s}^2$$

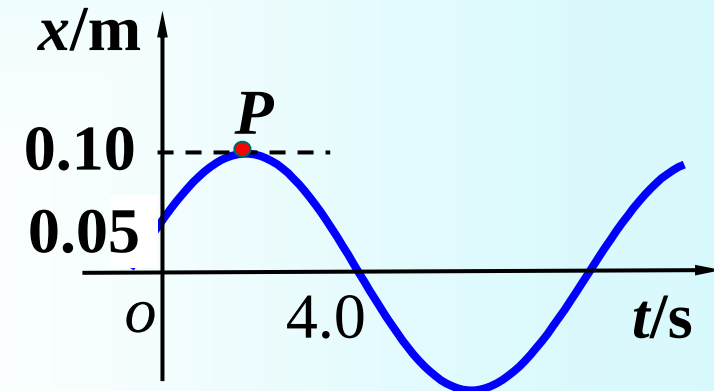
Example 2 Figure gives the displacements $x(t)$ of a particle undergoing SHM. Find (a) the motion equation (运动方程) and (b) the time interval to reach point P.

Solution: we can find the phase from the curve

(a) $t = 0, \quad \varphi = -\frac{\pi}{3} \quad t = 4\text{s}, \quad \omega t + \varphi = \frac{\pi}{2}$

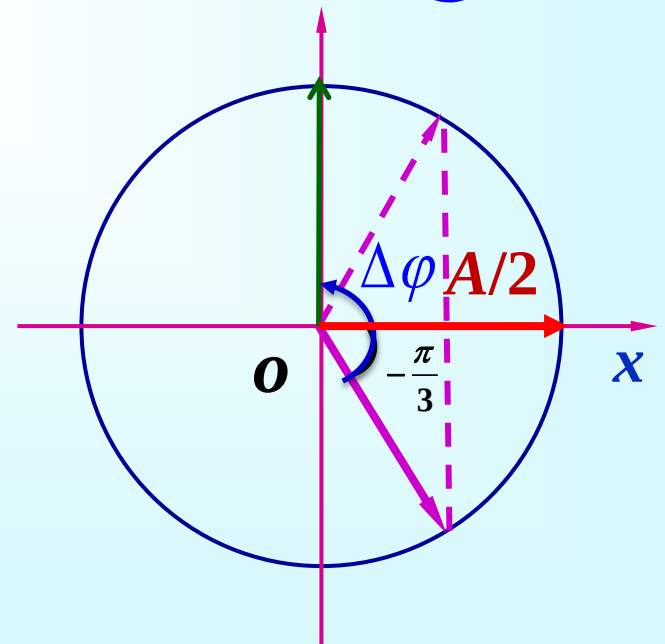
$$\omega = \frac{\Delta\varphi}{\Delta t} = \frac{\frac{1}{2}\pi - (-\frac{1}{3}\pi)}{4} = \frac{5}{24}\pi \text{ s}^{-1}$$

$$x = 0.1 \cos\left(\frac{5}{24}\pi t - \frac{1}{3}\pi\right) \text{ m}$$

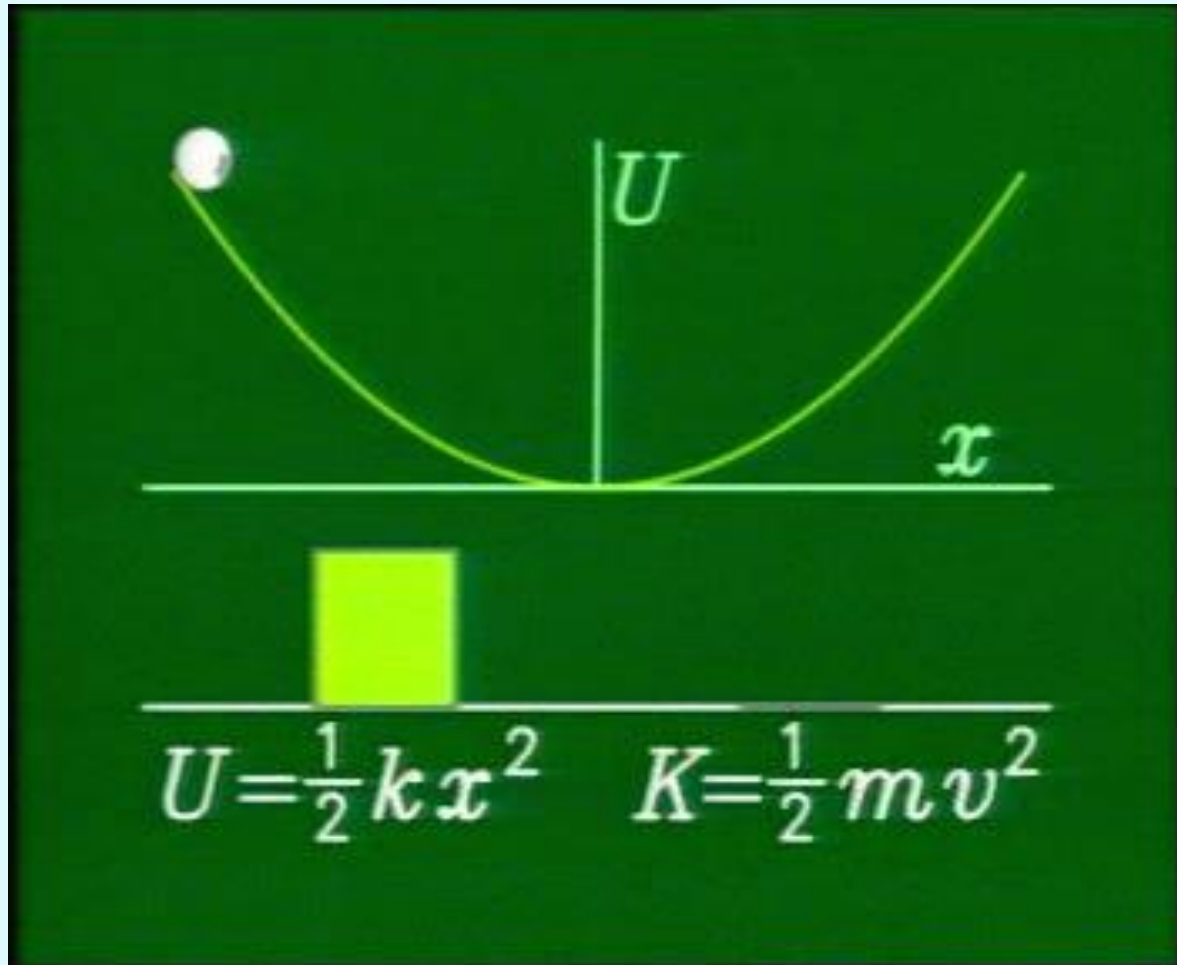


(b) Phase of point P: 0

$$\Delta t = \frac{\Delta\varphi}{\omega} = \frac{0 - (-\frac{1}{3}\pi)}{5\pi / 24} = 1.6 \text{ s}$$



16.3 Energy in SHM



For SHO, energy $E = U + K$

$$x(t) = A \cos(\omega t + \varphi) \qquad v = -A\omega \sin(\omega t + \varphi)$$

Potential energy

$$U = \frac{1}{2} kx^2 = \frac{1}{2} kA^2 \cos^2(\omega t + \varphi)$$

$$U_{\max} = \frac{1}{2} kA^2, \quad U_{\min} = 0$$

$$\bar{U} = \frac{1}{T} \int_0^T \frac{1}{2} kA^2 \cos^2(\omega t + \varphi) dt$$

$$= \frac{1}{T} \frac{1}{2} kA^2 \int_0^T \frac{1 + \cos 2(\omega t + \varphi)}{2} dt = \frac{1}{T} \frac{1}{2} kA^2 \left(\frac{T}{2} + 0 \right) = \frac{1}{4} kA^2$$

Kinetic energy

$$K = \frac{1}{2} m v^2 = \frac{1}{2} m \omega^2 A^2 \sin^2 (\omega t + \varphi)$$

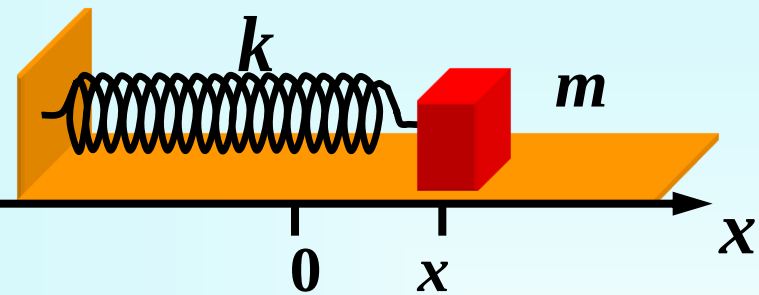
$$m \omega^2 = k$$

$$= \frac{1}{2} k A^2 \sin^2 (\omega t + \varphi)$$

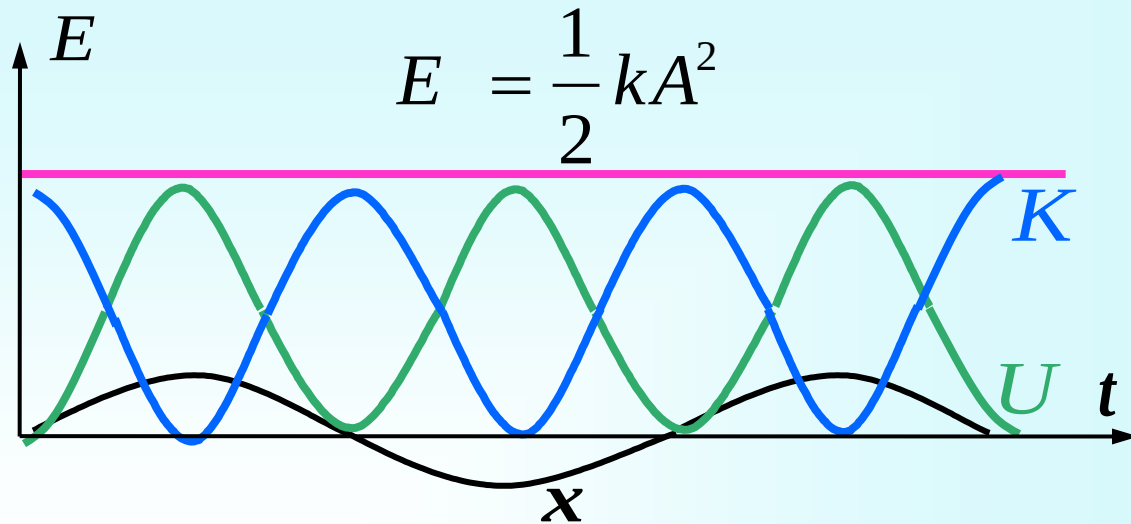
$$K_{\max} = \frac{1}{2} k A^2, K_{\min} = 0$$

$$\bar{K} = \frac{1}{T} \int_0^T \frac{1}{2} k A^2 \sin^2 (\omega t + \varphi) dt$$

$$= \frac{1}{T} \times \frac{1}{2} k A^2 \int_0^T \frac{1 - \cos 2(\omega t + \varphi)}{2} dt = \frac{1}{4} k A^2$$



$$E = K + U = \frac{1}{2} k A^2$$



(1) The total mechanical energy of the SHO is conserved and is proportional to the **square** of the amplitude $E \propto A^2$. **Potential** energy is the **maximum**, while **kinetic energy** is **zero**.

(2) Both potential energy and kinetic energy vary with the same frequency **$2f$** .

$$(3) \bar{U} = \bar{K} = \frac{1}{4} k A^2 = \frac{1}{2} E$$

➤ Conservation of mechanical energy

 motion equation

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \text{constant}$$

$$\frac{d}{dt} \left(\frac{1}{2}mv^2 + \frac{1}{2}kx^2 \right) = 0$$

$$\cancel{mv} \frac{dv}{dt} + kx \frac{\cancel{dx}}{dt} = 0$$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

动力学方程

* The total energy U present at any instant in an oscillating LC circuit is given by

$$U = U_B + U_E = \frac{Li^2}{2} + \frac{q^2}{2C}$$

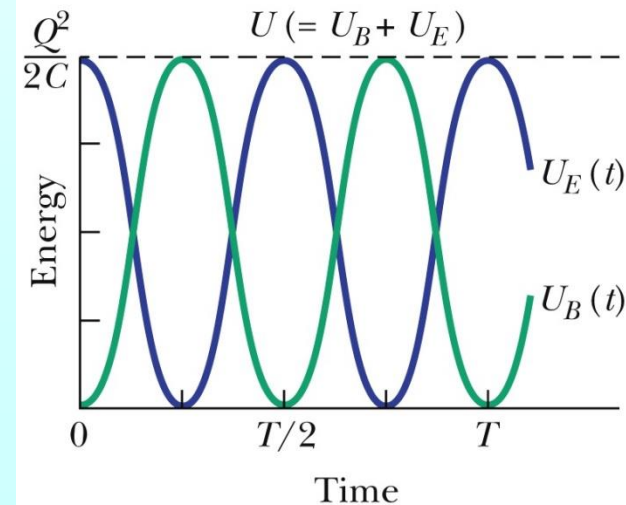
$$\frac{dU}{dt} = \frac{d}{dt} \left(\frac{Li^2}{2} + \frac{q^2}{2C} \right) = Li \frac{di}{dt} + \frac{q}{C} \frac{dq}{dt} = 0.$$

$$i = \frac{dq}{dt} \text{ and } \frac{di}{dt} = \frac{d^2q}{dt^2}.$$

$$\frac{d^2q}{dt^2} + \frac{1}{LC}q = 0$$

$$\omega = \frac{1}{\sqrt{LC}} \quad (LC \text{ circuit}).$$

The electrical and magnetic energies vary but the total is constant.



Question

The block has a kinetic energy of 3J and the spring has an elastic potential energy of 2J when the block is at $x=+20\text{cm}$.

(a) What is the kinetic energy when the block is at $x=0$ **5J**

What is the elastic potential energy when the block is at

(b) $x= -20\text{cm}$ and (c) $x= -x_m$

2J

5J

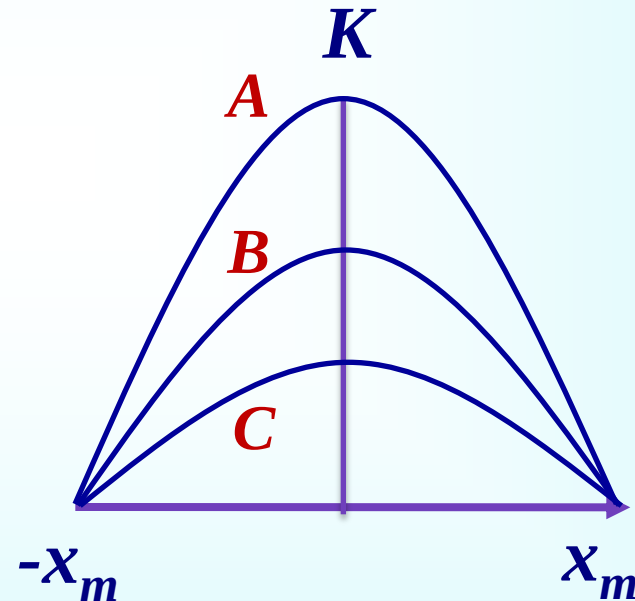
Question

Figure shows plots of the kinetic energy K versus position x for three harmonic oscillators that have the **same mass**.

Rank the plots according to (a) the corresponding **spring constant** and (b) the corresponding **period** of the oscillator, greatest first.

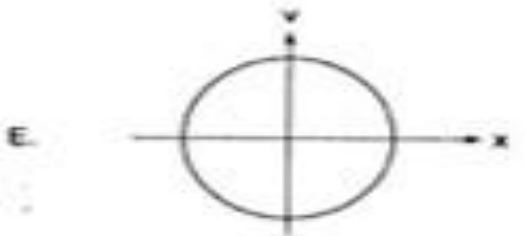
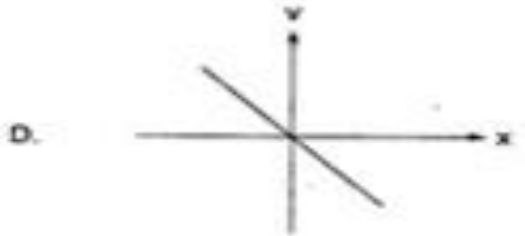
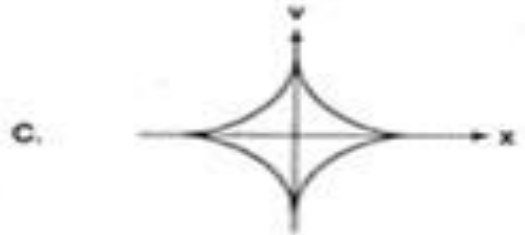
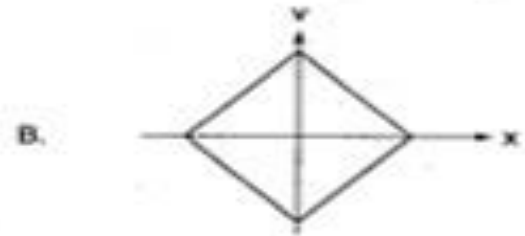
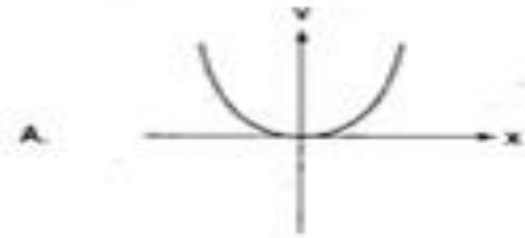
(a) $A > B > C$

(b) $A < B < C$



Question

Which of the following graphs shows the relationship between **the displacement** and **velocity** of a mass oscillating on a spring?



Question

Five graphs of a force as a function of position are shown in figure. In **how many** cases can an object subject to this force undergo simple harmonic motion?

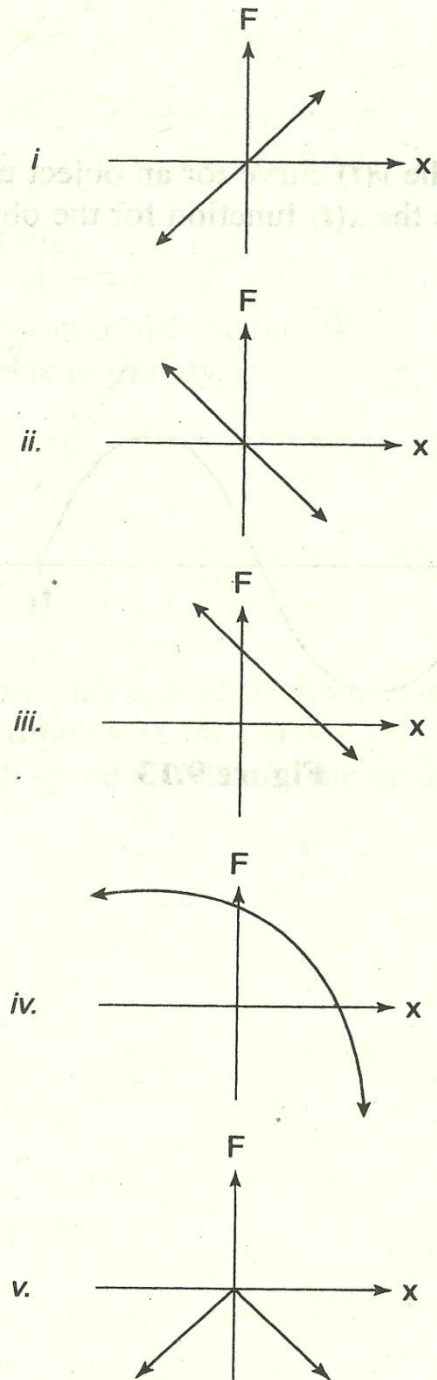
(A) none

(B) one

(C) two

(D) three

(E) four



Solving problems **Determine motion equation** ω, A, φ

1. determine ω

$$\omega = \sqrt{\frac{k}{m}}, \omega = \sqrt{\frac{g}{l}}, \omega = \sqrt{\frac{mgl}{I}} \quad \text{or} \quad \omega = \frac{2\pi}{T} \quad \text{From curve } x(t)$$

2. determine A

if x_0, v_0 are known $\Rightarrow A, \varphi, x(t)$

$$x_0 = A \cos \varphi \quad v_0 = -A\omega \sin \varphi$$

$$A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}}$$

$$\phi = \tan^{-1}\left(-\frac{v_0}{\omega x_0}\right)$$

$$A = \sqrt{\frac{2E}{k}} = \sqrt{\frac{2E_0}{k}}$$

$$E \propto A^2$$

3. Determine φ

$$x(0) = A \cos \varphi$$

$$|\cos \varphi| \Rightarrow \varphi = \{\varphi, -\varphi\}$$

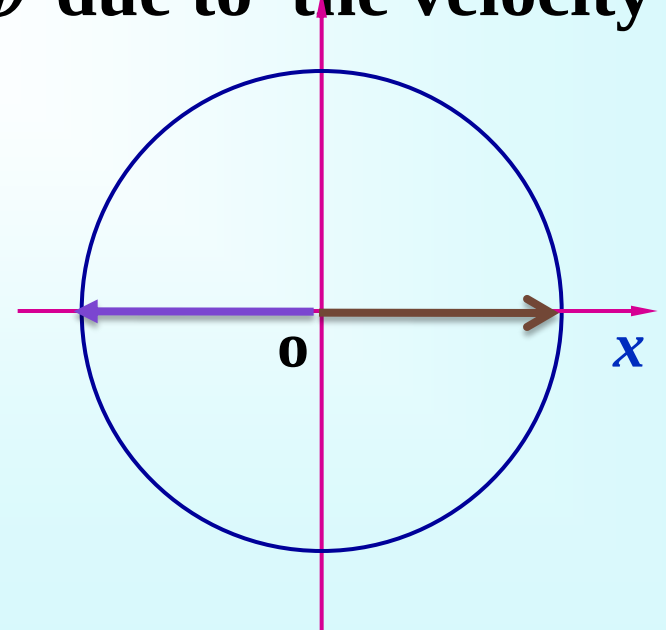
(a) We choose the correct sign of φ from rotation vector.

(b) We choose the correct sign of φ due to the velocity .

$$v = -A\omega \sin(\omega t + \varphi)$$

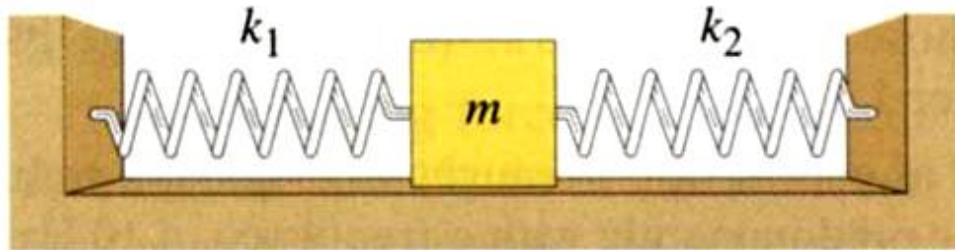
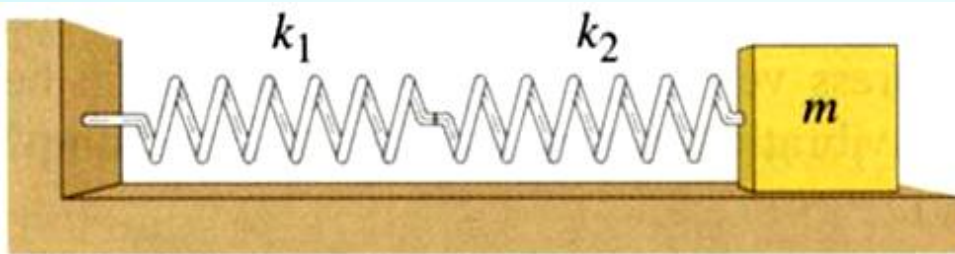
Special cases: $x_0 = A \rightarrow \varphi = 0$

$$x_0 = -A \rightarrow \varphi = \pi$$

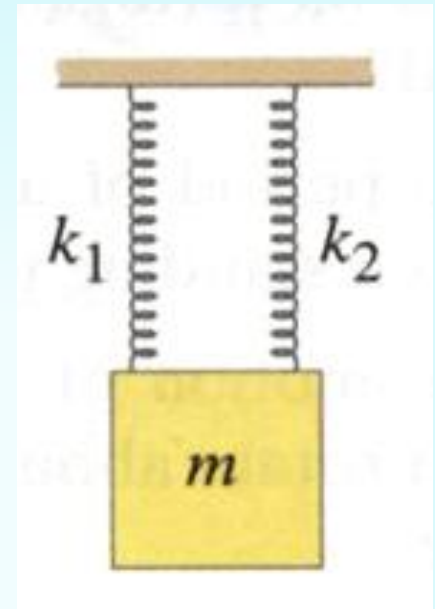


Examples

$$T = 2\pi \sqrt{m \left(\frac{1}{k_1} + \frac{1}{k_2} \right)}$$



$$T = 2\pi \sqrt{\frac{m}{k_1 + k_2}}$$



$$T = 2\pi \sqrt{\frac{m}{k_1 + k_2}}$$

Example 4

A 450-g object oscillates from a vertically hanging light spring once every 0.5s. (a) Write down the equation giving its position y (+upward) as a function of time t , assuming it started by being compressed 10cm from the equilibrium position (where $x=0$), and released. (b) How long will it take to get to the equilibrium position for the first time? (c) What will be its maximum speed? (d) What will be its maximum acceleration, and when will it first be attained?

Solution:

➤ Prove: the motion is SHM.

+ : upward

Equilibrium position

$$mg = kx_0 \quad (x_0 > 0)$$

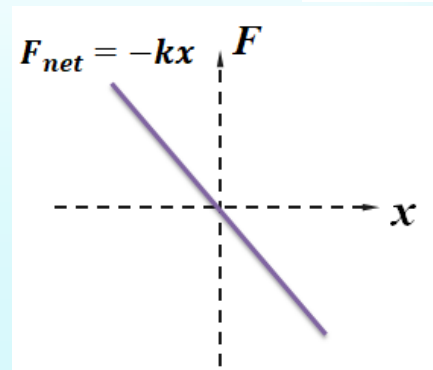
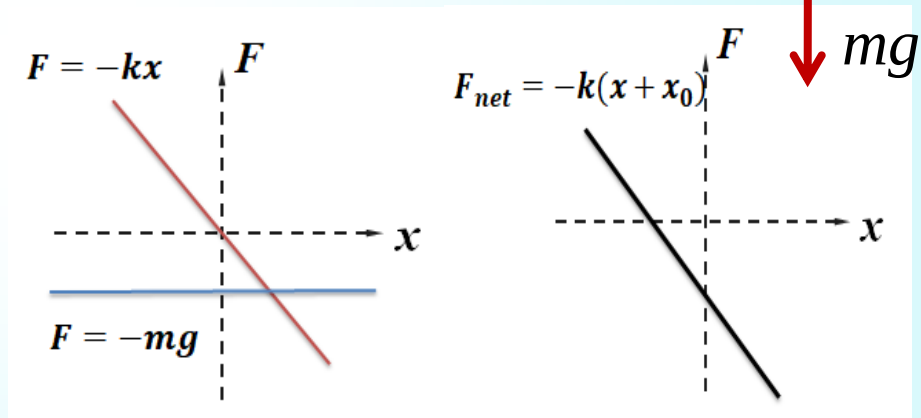
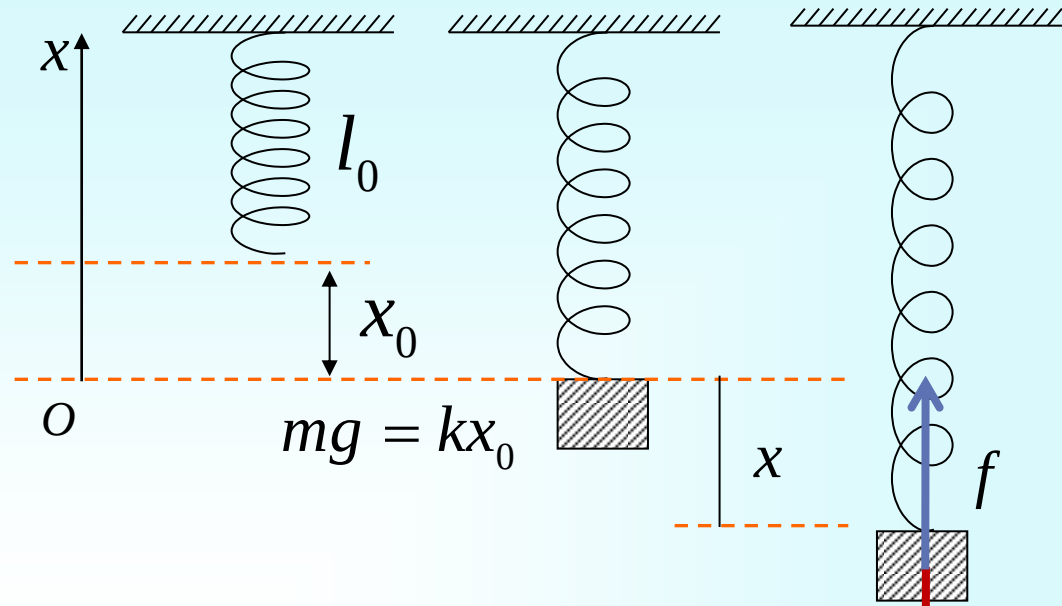
At position x

$$F_{\text{net}} = -k(x - x_0) - mg = -kx$$

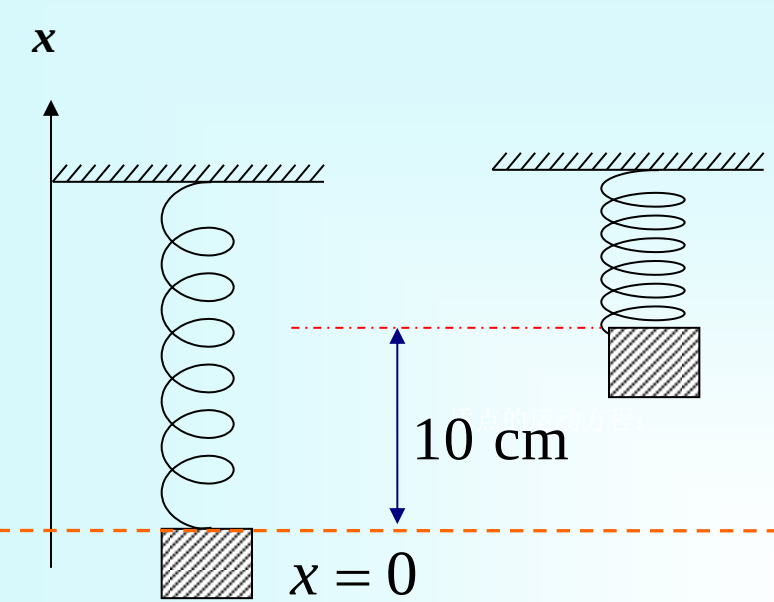
$$m \frac{d^2 x}{dt^2} = -kx \Rightarrow \frac{d^2 x}{dt^2} + \frac{k}{m} x = 0$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$x = A \cos(\omega t + \varphi)$$



以平衡位置为坐标原点



Solution:

$$(a) A = 0.1 \text{ m}, T = 0.5 \text{ s}$$

$$\varphi = 0$$

$$\omega = \frac{2\pi}{T} = 4\pi \text{ rad/s}$$

$$y = 0.1 \cos 4\pi t \text{ m}$$

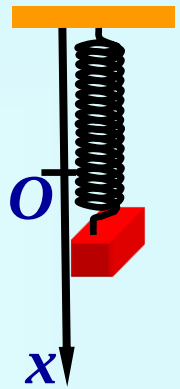
$$(b) \Delta t = \frac{T}{4} = 0.125 \text{ s}$$

$$(c) v_{\max} = A\omega = 0.1 \times 4\pi = 0.4\pi \text{ m/s}$$

$$(d) a_{\max} = A\omega^2 = 1.6\pi^2 \text{ m/s}^2$$

$$\Delta t' = \frac{1}{2} T = 0.25 \text{ s}$$

Example 5 A mass m is gently placed on the end of a freely hanging spring. The mass then falls 22.0 cm before it stops and begins to rise. What is the **frequency** of the oscillation?



Solution: $A = \frac{h}{2} = \frac{0.22}{2} = 0.11\text{m}$ $mg = kA \Rightarrow k = 2mg / h$

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{y_0}} = 9.44 \text{ s}^{-1} \qquad f = \frac{\omega}{2\pi} = 1.50 \text{ Hz}$$

Or we use **energy conservation** between the initial point and the lowest point

$$K_i + U_i = K_f + U_f \qquad 0 + mgh = 0 + \frac{1}{2}kh^2 \Rightarrow k = 2mg / h$$

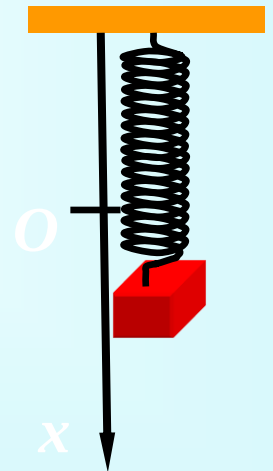
$$f = \frac{\omega}{2\pi} = \sqrt{\frac{k}{m}} / 2\pi = \sqrt{\frac{2g}{h}} / 2\pi = 1.50 \text{ Hz}$$

Question A mass m hanging on a spring oscillates vertically, If the equilibrium point of the oscillation is a distance d below the relaxed length of the spring and if the amplitude of the oscillation is A , what is the **maximum kinetic energy** of the oscillation?

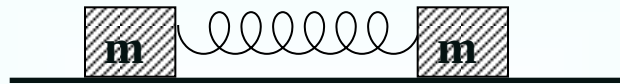
(A) $\frac{1}{2}(mg/d)d^2$ (B) $\frac{1}{2}(mg/d)A^2$

(C) $\frac{1}{2}(mg/d)A^2 + mgd$

(D) $\frac{1}{2}(mg/d)A^2 - mgd$



Example 6 Two equal masses m lying on a **frictionless** horizontal surface are attached to each other by a spring of spring constant k as shown. The spring is compressed a distance Δx , and then both masses are released from rest at the same instant.

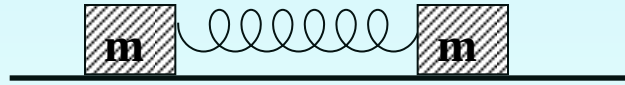


- (a) What is the **maximum speed** of masses?
- (b) What is the **period** of oscillation of each mass?

The same experiment is then performed with the difference that the two blocks attached to the spring have different masses (m and M).

- (c) What is the **maximum speed** of mass M ?
- (d) What is the **amplitude** of oscillation of mass M ?

Solution:



$$(a) \frac{1}{2} k (\Delta x)^2 = 2 \left(\frac{1}{2} m v^2 \right) \qquad v = \sqrt{\frac{k (\Delta x)^2}{2m}}$$

$$(b) F = -k_{\text{effective}} x \qquad T = 2\pi \sqrt{m / k_{\text{eff}}} = 2\pi \sqrt{m / 2k}$$

$$(c) \frac{1}{2} k (\Delta x)^2 = \frac{1}{2} m v_m^2 + \frac{1}{2} M v_M^2 \qquad m v_m = M v_M$$

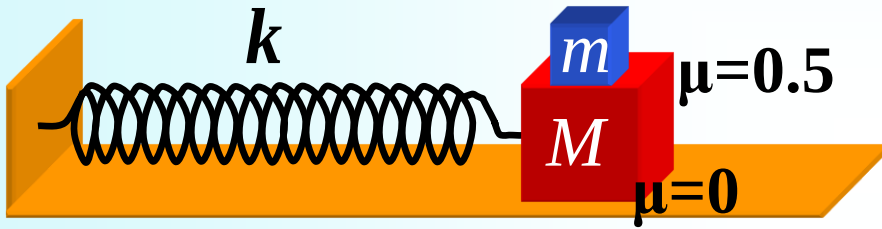
$$v = \sqrt{\frac{k (\Delta x)^2}{M (1 + M / m)}}$$

$$(d) 0 = \frac{M \Delta x_M - m \Delta x_m}{m + M} \qquad \Delta x = \Delta x_M + \Delta x_m \qquad \Delta x_M = \frac{\Delta x}{1 + M / m}$$

Example 7

This is a SHO . In order to keep the block m on the big block M , what is the **maximum energy** that the system could has.

$$M = 0.6kg, m = 0.4kg, k = 25 N / m, \mu_s = 0.5$$



Solution: $kA = (m + M)a$

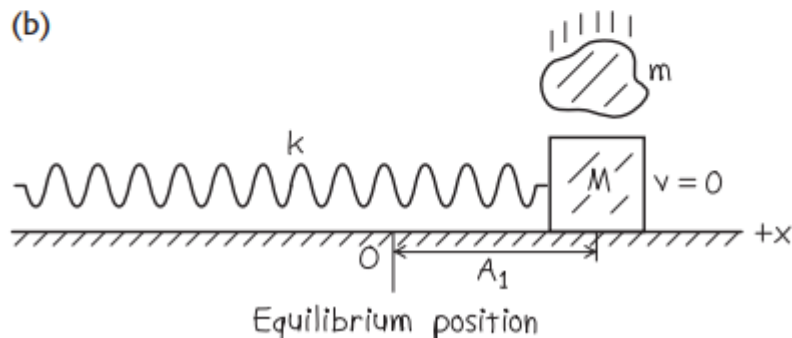
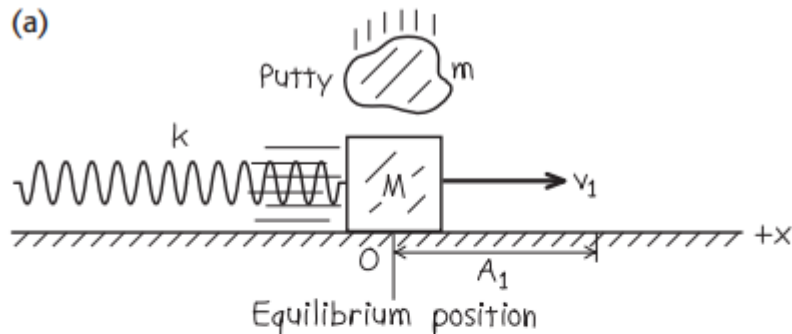
$$mg\mu = ma$$

$$A = \frac{(m + M)}{k} g\mu = \frac{0.6 + 0.4}{25} \times 10 \times 0.5 = 0.2m$$

$$E = \frac{1}{2} kA^2 = 0.5 \times 25 \times 0.2^2 = 0.5J$$

Question

A block of mass M attached to a horizontal spring with force constant k is moving in SHM with amplitude A_1 . As the block passes through its **equilibrium position**, a lump of putty of mass m is dropped from a small height and **sticks** to it. (a) Find the new **amplitude** and **period** of the motion. (b) Repeat part (a) if the putty is dropped onto the block when it is at one **end** of its path.



$$A_2 = A_1 \sqrt{\frac{M}{M+m}}$$

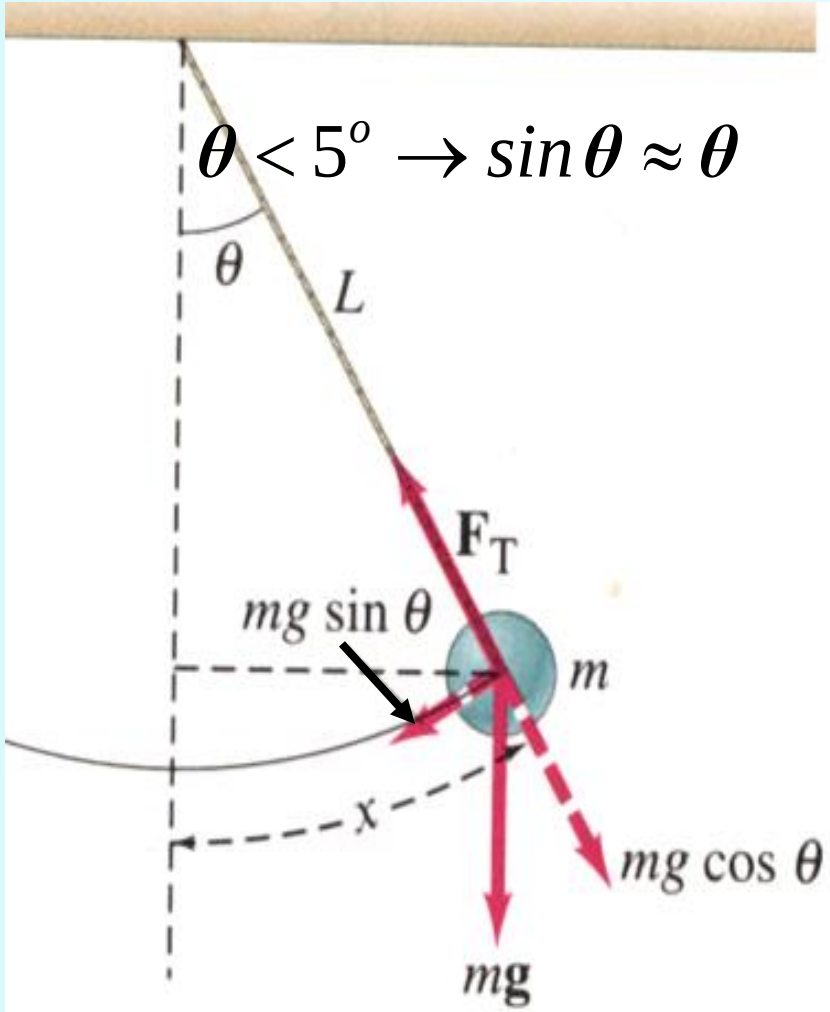
$$T_2 = 2\pi \sqrt{\frac{M+m}{k}}$$

$$A_2 = A_1$$

$$T_2 = 2\pi \sqrt{\frac{M+m}{k}}$$

16.4 Pendulums

1.Simple pendulum



Ignore mass of string and any resistance.

$$F = -mg \sin \theta = m \frac{dv}{dt} = \frac{d^2 \theta}{dt^2}$$

$$\frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0$$

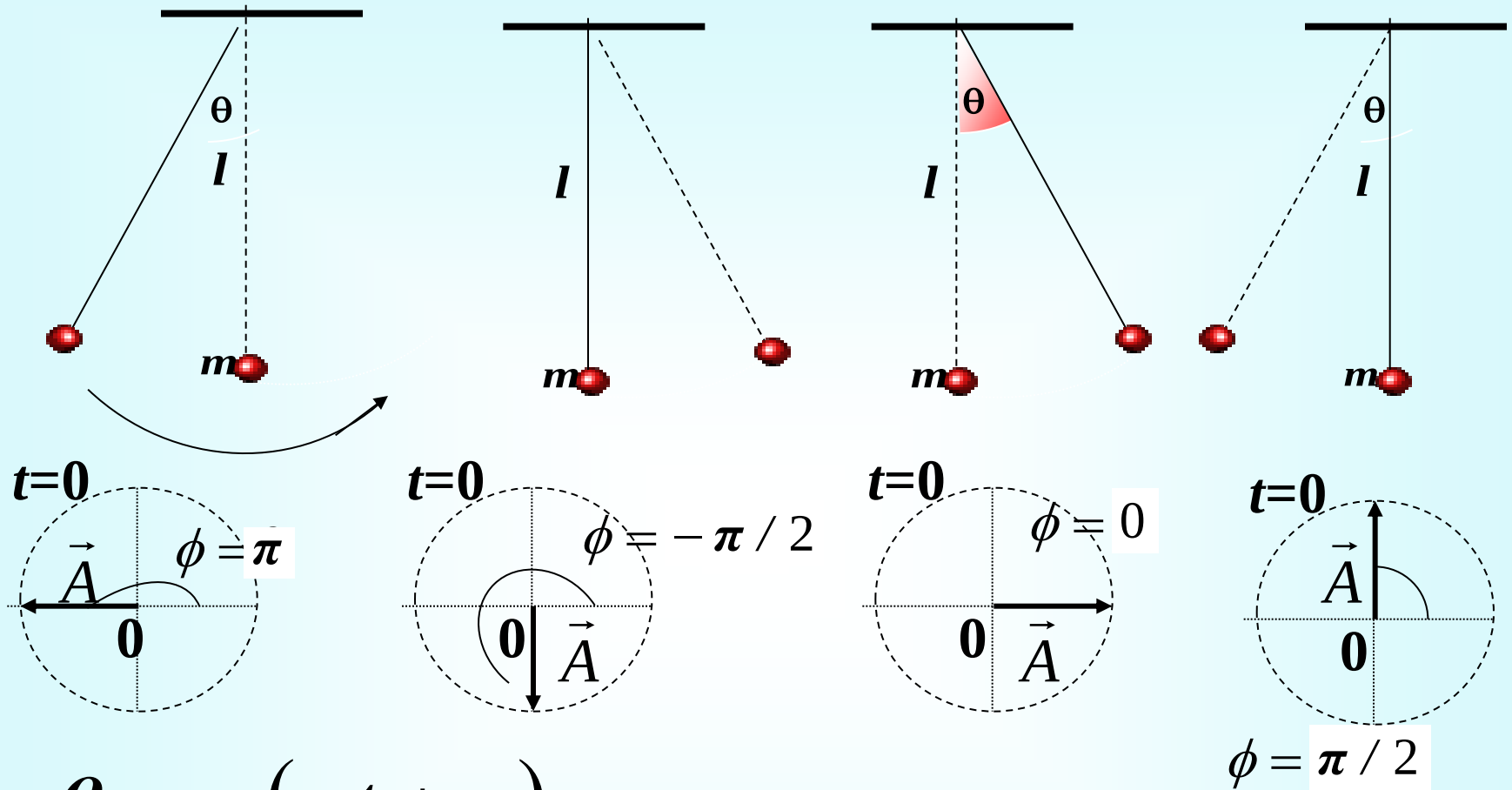
$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$\theta = \theta_m \cos(\omega t + \varphi)$$

$$\omega' = \frac{d\theta}{dt} = -\omega \theta_m \sin(\omega t + \varphi)$$

$$\alpha = \frac{d\omega'}{dt} = -\omega^2 \theta_m \cos(\omega t + \varphi)$$

Determine phase angle



$$\theta = \theta_0 \cos(\omega t + \pi)$$

$$\theta = \theta_0 \cos(\omega t - \pi / 2)$$

$$\theta = \theta_0 \cos(\omega t + \pi / 2)$$

$$\theta = \theta_0 \cos(\omega t)$$

2. Physical pendulum ($\theta < 5^\circ$)

$$\sum \tau = I\alpha = I \frac{d^2\theta}{dt^2}$$

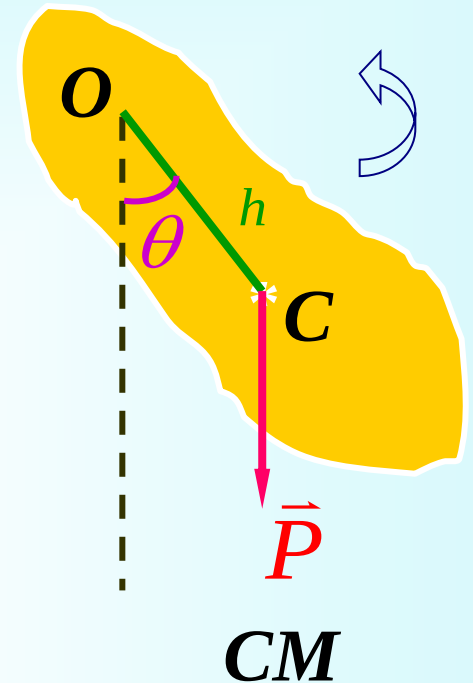
$$I \frac{d^2\theta}{dt^2} = -mgh \sin \theta$$

$$\frac{d^2\theta}{dt^2} + \omega^2 \theta = 0$$

If $\omega^2 = \frac{mgh}{I}$ $T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{I}{mgh}}$

$$\theta = \theta_m \cos(\omega t + \phi)$$

Angular SHM 角谐振动



3. Torsion pendulum

Restoring torque

$$\tau = -\kappa\theta$$

κ : torsion constant (depends on the length, diameter and material of the suspension wire)

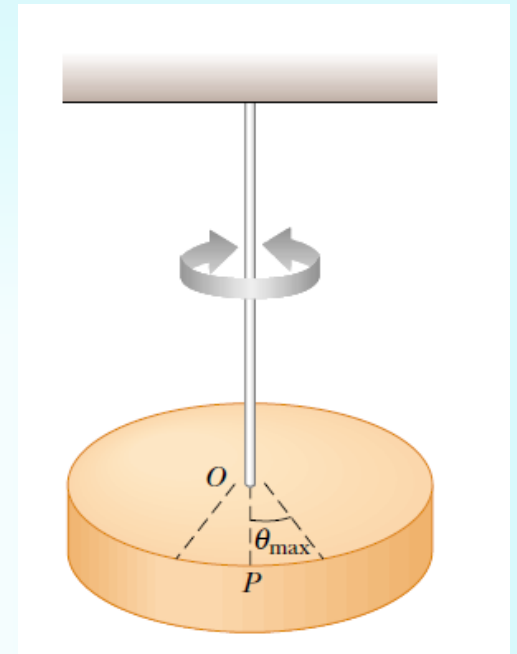
Comparison with

$$F = -k_{\text{effective}}x \qquad T = 2\pi\sqrt{\frac{m}{k}}$$

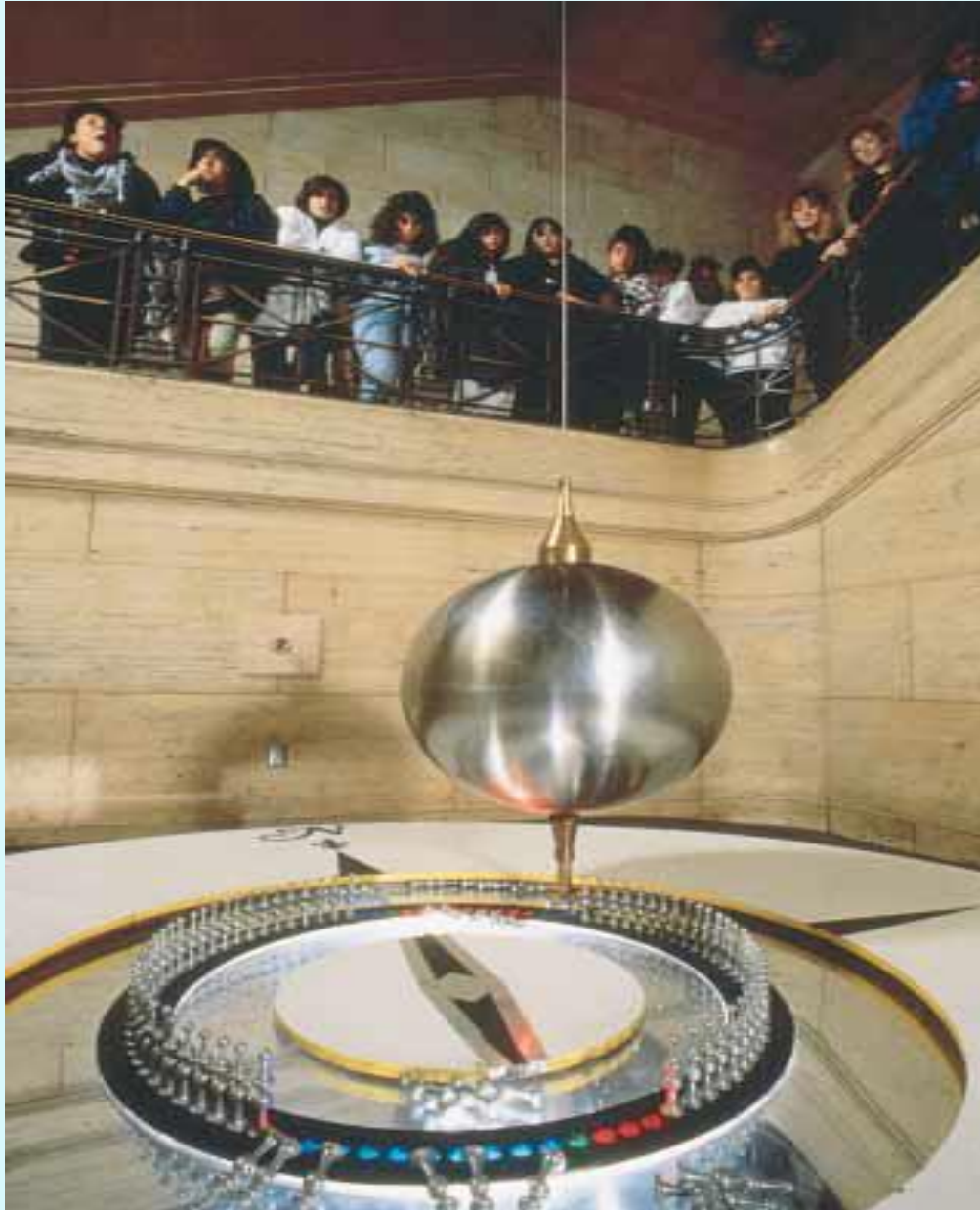
We replace the spring constant **k** and mass **m** with its equivalents **κ** and **I** .

$$T = 2\pi\sqrt{\frac{I}{\kappa}}$$

Angular SHM 角谐振动



Foucault pendulum



The **Foucault pendulum** at the Franklin Institute in Philadelphia. This type of pendulum was first used by the French physicist Jean Foucault to verify the **Earth's rotation** experimentally. As the pendulum swings, the vertical plane in which it oscillates appears to rotate as the bob successively knocks over the indicators arranged in a circle on the floor. In reality, the plane of oscillation is fixed in space, and the Earth rotating beneath the swinging pendulum moves the indicators into position to be knocked down, one after the other.

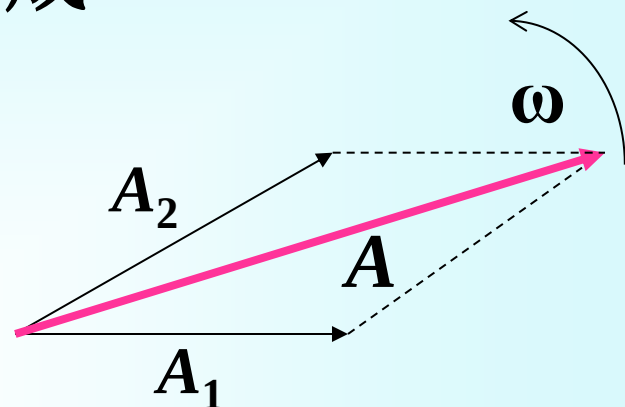
16.5 Composition of SHM

1. 同方向同频率的简谐振动的合成

► 解析法

(1) 分振动： $x_1 = A_1 \cos(\omega t + \varphi_1)$

$$x_2 = A_2 \cos(\omega t + \varphi_2)$$



(2) 合振动： $x = x_1 + x_2$

$$x = A_1 \cos(\omega t + \varphi_1) + A_2 \cos(\omega t + \varphi_2)$$

$$= A_1(\cos \omega t \cos \varphi_1 - \sin \omega t \sin \varphi_1) + A_2(\cos \omega t \cos \varphi_2 - \sin \omega t \sin \varphi_2)$$

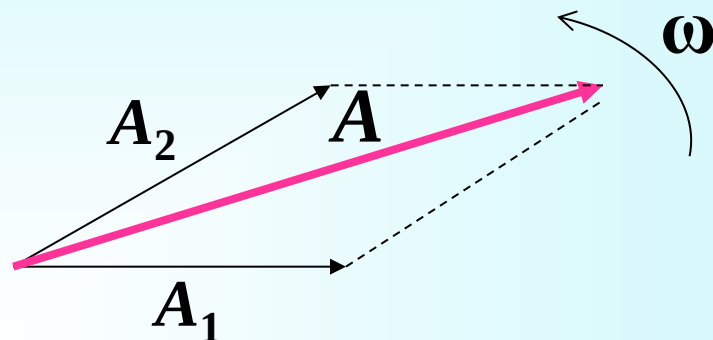
$$= (A_1 \cos \varphi_1 + A_2 \cos \varphi_2) \cos \omega t - (A_1 \sin \varphi_1 + A_2 \sin \varphi_2) \sin \omega t$$

$$= (A_1 \cos \varphi_1 + A_2 \cos \varphi_2) \cos \omega t - (A_1 \sin \varphi_1 + A_2 \sin \varphi_2) \sin \omega t$$

引入 A, ϕ

$$A_1 \cos \varphi_1 + A_2 \cos \varphi_2 = A \cos \varphi$$

$$A_1 \sin \varphi_1 + A_2 \sin \varphi_2 = A \sin \varphi$$



$$x = A \cos \varphi \cos \omega t - A \sin \varphi \sin \omega t = A \cos(\omega t + \varphi)$$

合振动是简谐振动，其频率仍为 ω

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\varphi_2 - \varphi_1)}$$

$$\text{tg } \varphi = \frac{A_1 \sin \varphi_1 + A_2 \sin \varphi_2}{A_1 \cos \varphi_1 + A_2 \cos \varphi_2} \quad \min(\varphi_1, \varphi_2) \leq \phi \leq \max(\varphi_1, \varphi_2)$$

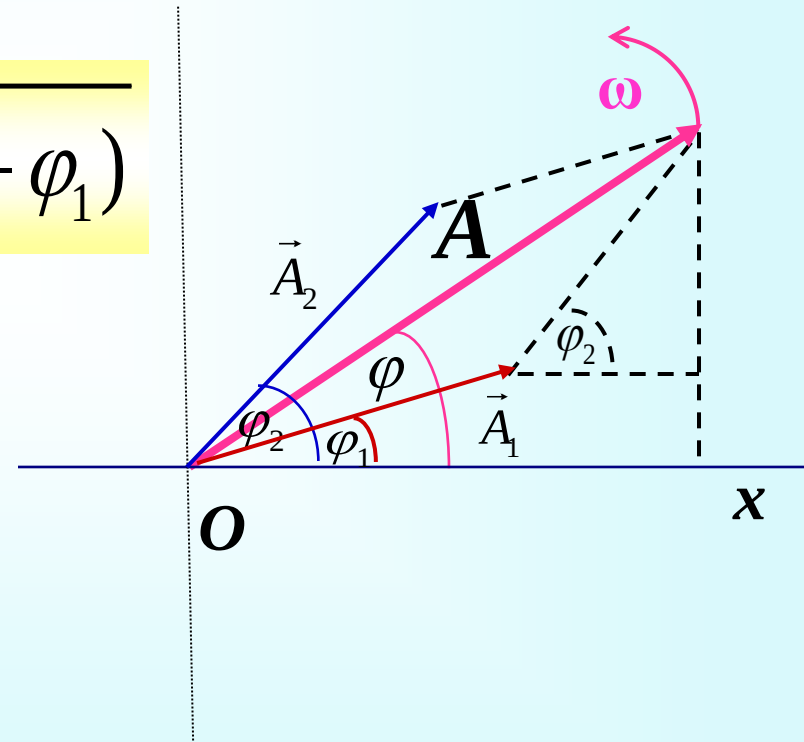
► 旋转矢量合成法

合振动： $x = x_1 + x_2$

$$x = A \cos(\omega t + \varphi)$$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\varphi_2 - \varphi_1)}$$

$$\tan \varphi = \frac{A_1 \sin \varphi_1 + A_2 \sin \varphi_2}{A_1 \cos \varphi_1 + A_2 \cos \varphi_2}$$



➤相位差: $\Delta\varphi = (\omega t + \varphi_2) - (\omega t + \varphi_1) = \varphi_2 - \varphi_1$

对两同频率的谐振动 $\Delta\varphi = \varphi_2 - \varphi_1$ (初相差)

两种特殊情况

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\varphi_2 - \varphi_1)}$$

(a) 若 $\varphi_2 - \varphi_1 = \pm 2k\pi$ ($k = 0, 1, 2 \dots$)

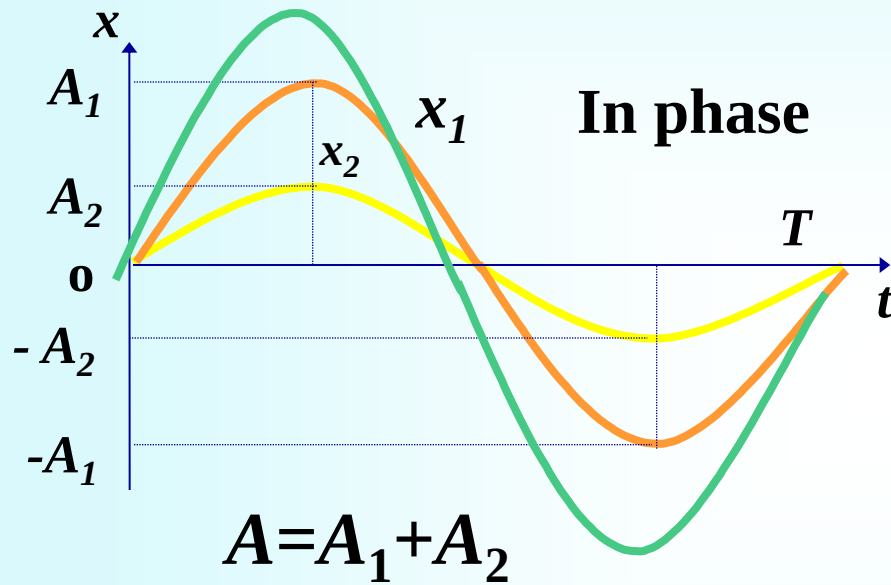
则 $A = A_1 + A_2$, 两分振动相互加强

(b) 若 $\varphi_2 - \varphi_1 = \pm(2k + 1)\pi$ ($k = 0, 1, 2 \dots$)

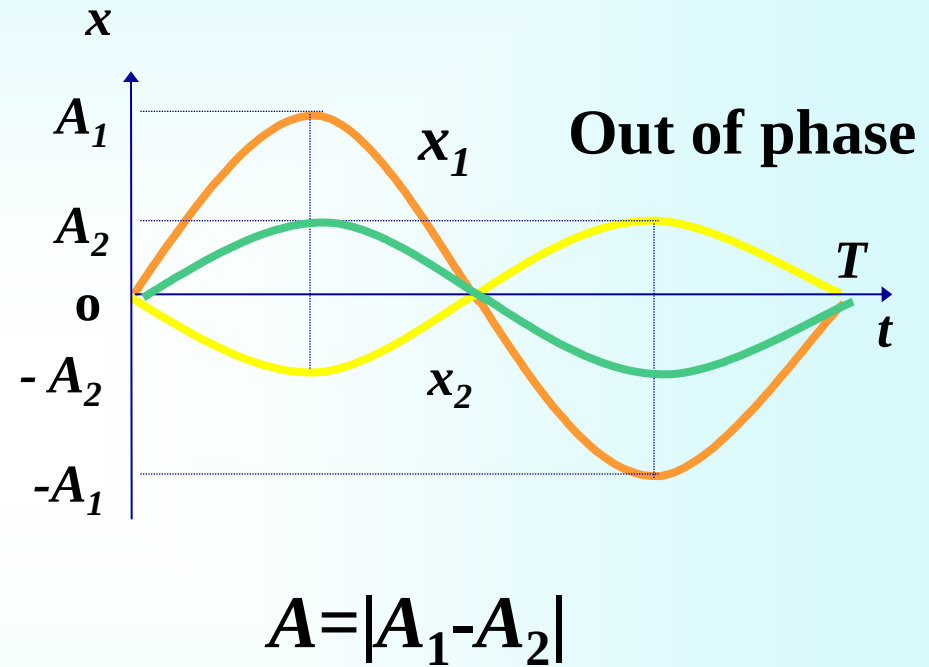
则 $A = |A_1 - A_2|$, 两分振动相互减弱

如 $A_1 = A_2$, 则 $A = 0$

$$\varphi_2 - \varphi_1 = \pm 2k\pi \quad (k = 0, 1, 2 \dots)$$



$$\varphi_2 - \varphi_1 = \pm (2k + 1)\pi \quad (k = 0, 1, 2 \dots)$$



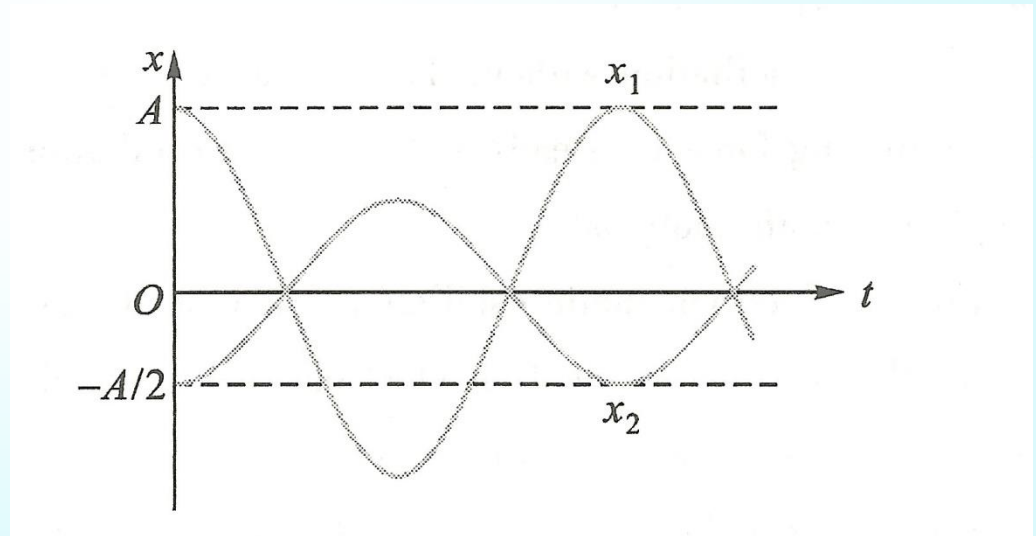
The figure shows the curves of two simple harmonic motions. If the two simple harmonic motions can be superimposed, the phase constant of the resultant cosine oscillation(余弦函数表示) is

(A) $\frac{3}{2}\pi$

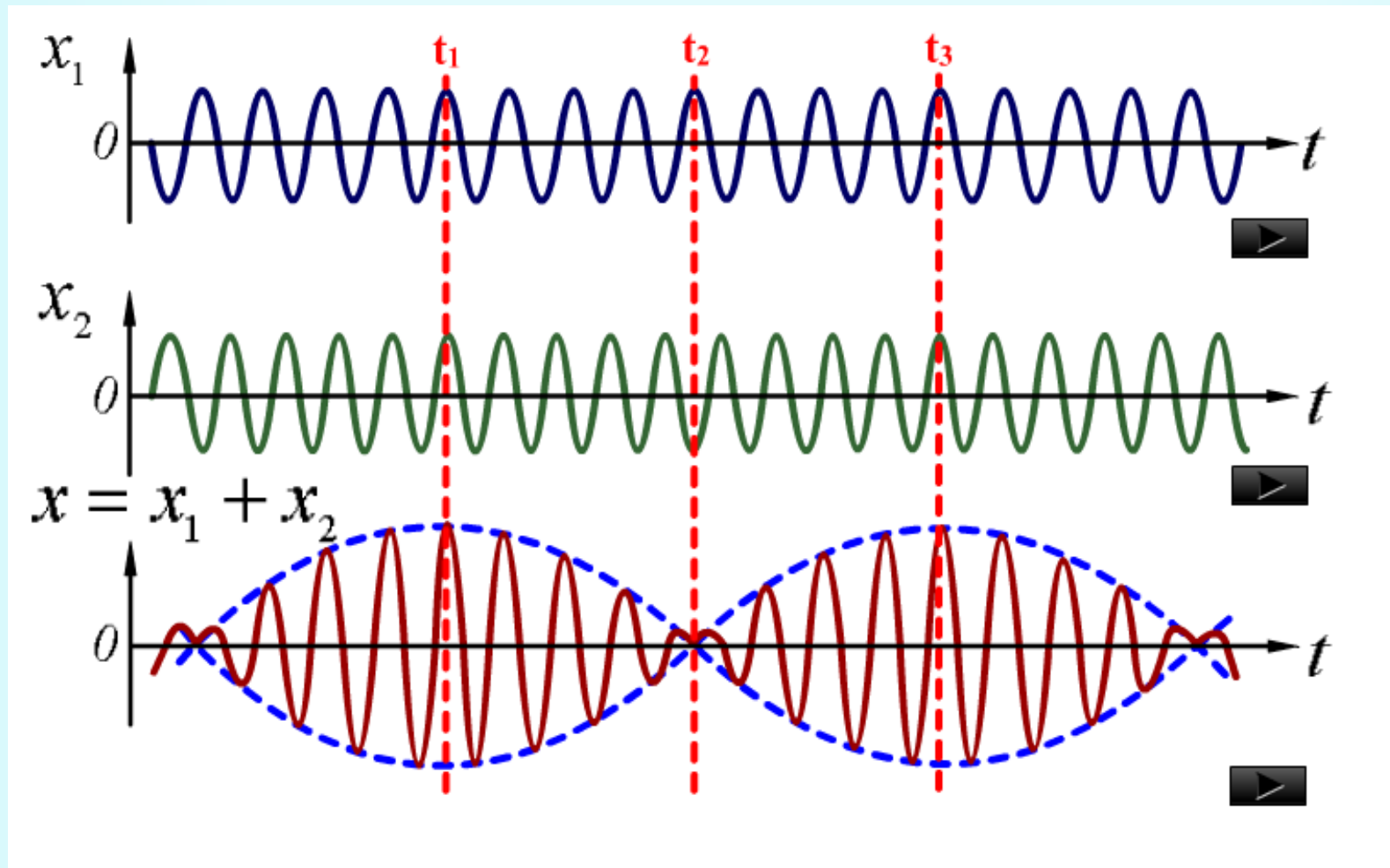
(B) $\frac{1}{2}\pi$

(C) π

(D) 0



2. 两个同方向不同频率的简谐振动的合成



➤解析法

分振动： $x_1 = A \cos \omega_1 t$

$x_2 = A \cos \omega_2 t$

同振幅，同初相

合振动： $x = x_1 + x_2$

$$= A \cos \omega_1 t + A \cos \omega_2 t = 2A \cos\left(\frac{\omega_1 + \omega_2}{2}t\right) \cos\left(\frac{\omega_1 - \omega_2}{2}t\right)$$

$$x = \left[2A \cos\left(\frac{\omega_1 - \omega_2}{2}t\right)\right] \cos\left(\frac{\omega_1 + \omega_2}{2}t\right)$$

随 t 缓变

随 t 快变

讨论 $|\omega_2 - \omega_1| \ll \omega_1 + \omega_2 \approx \omega_1 (\omega_2)$

$$x = A' \cos\left(\frac{\omega_1 + \omega_2}{2}t\right)$$

合振动不是简谐振动

频率较大而频率之差很小的两个同方向简谐运动的合成，其合振动的振幅时而加强时而减弱的现象叫拍。

$$x = \left[2A \cos 2\pi \left(\frac{f_1 - f_2}{2} \right) t \right] \cos 2\pi \left(\frac{f_1 + f_2}{2} \right) t$$

振幅部分

合振动频率

振动频率 $f = (f_1 + f_2)/2$

振幅 $A' = \left| 2A \cos 2\pi \left(\frac{f_2 - f_1}{2} \right) t \right|$

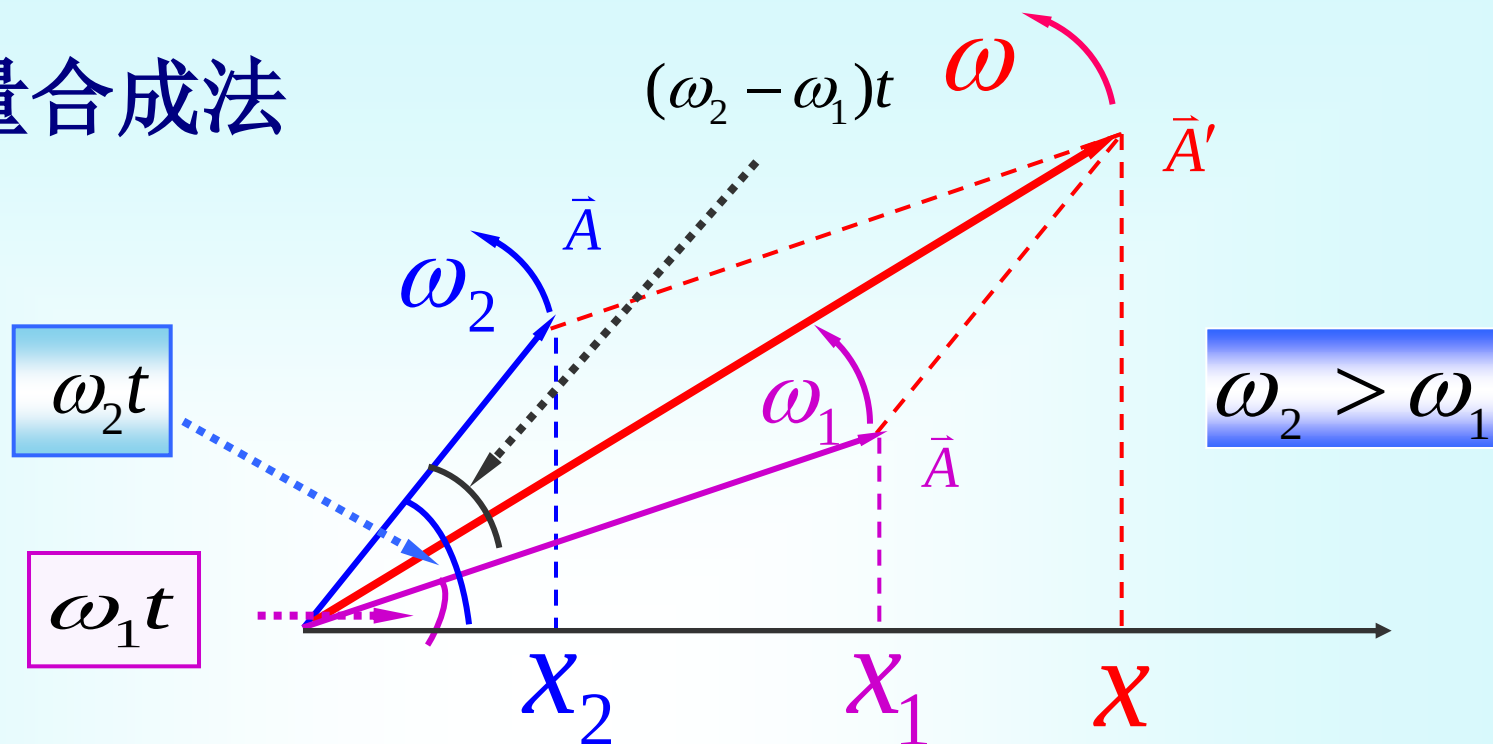
$$A_{\max} = 2A$$

$$A_{\min} = 0$$

$$f_{\text{beat}} = |f_1 - f_2|$$

拍频（振幅变化的频率）

➤ 旋转矢量合成法



振幅

$$A' = A \sqrt{2(1 + \cos \Delta\varphi)} = \left| 2A \cos \left[\left(\frac{\omega_2 - \omega_1}{2} \right) t \right] \right|$$

振动圆频率

$$\omega = \frac{\omega_1 + \omega_2}{2} \quad \leftarrow \quad \omega t = \frac{\omega_1 t + \omega_2 t}{2}$$

$$f_{beat} = |f_1 - f_2|$$

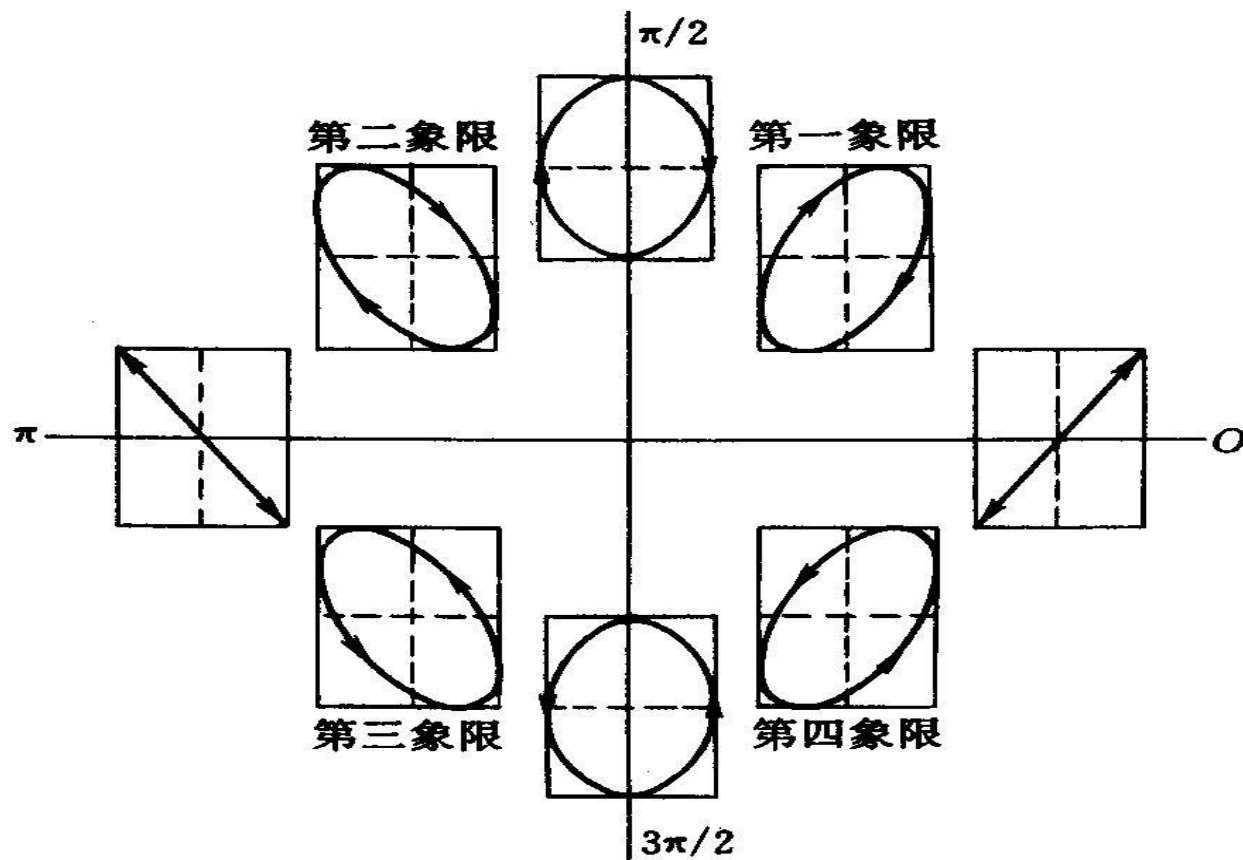


* 3. 两相互垂直同频率简谐振动的合成

$$x = A_1 \cos(\omega t + \varphi_1)$$

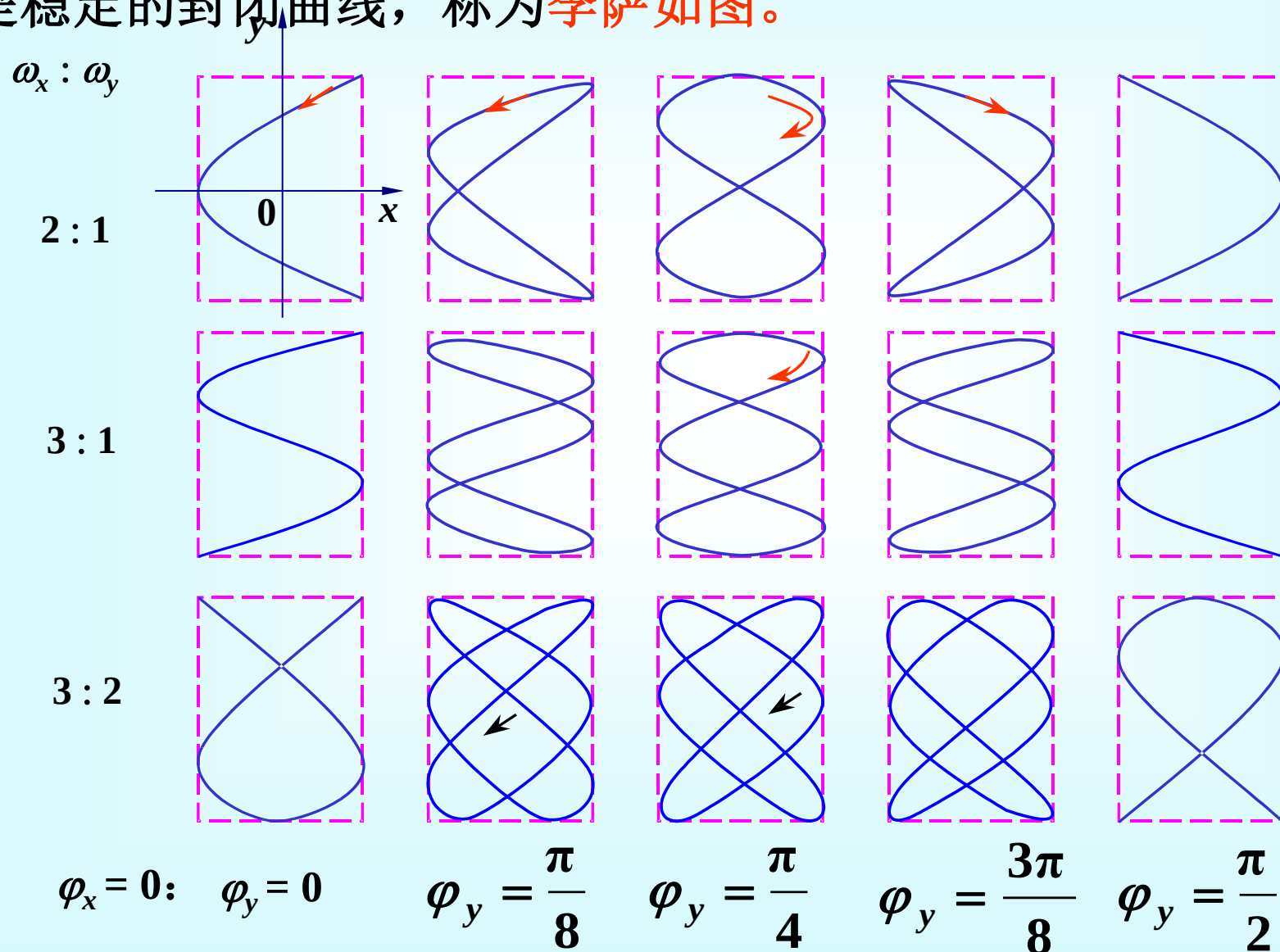
$$y = A_2 \cos(\omega t + \varphi_2)$$

$$\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} - \frac{2xy}{A_1 A_2} \cos(\varphi_2 - \varphi_1) = \sin^2(\varphi_2 - \varphi_1) \quad \text{—— 椭圆方程}$$



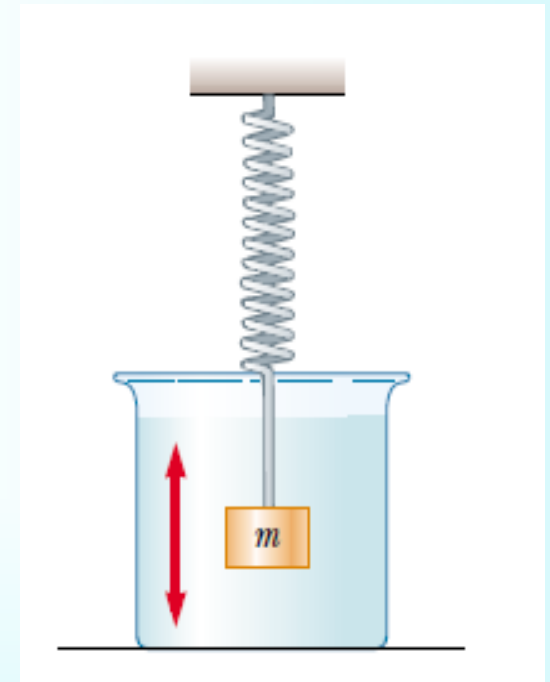
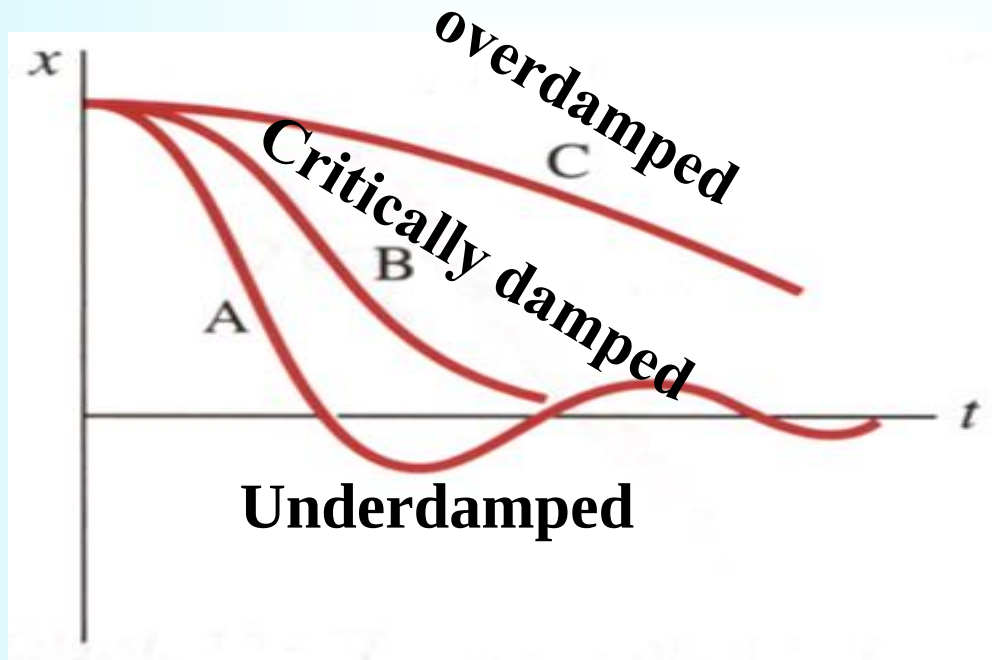
*4 互相垂直、不同频率简谐振动的合成

当两个分振动的频率比恰好等于简单的整数比时，合振动的轨迹是稳定的封闭曲线，称为李萨如图。



*16.6 Damped Harmonic Motion

Damped Harmonic Motion : the amplitude of any real oscillating slowly decreases with time until it stops.



Friction or viscosity are often proportional to velocity

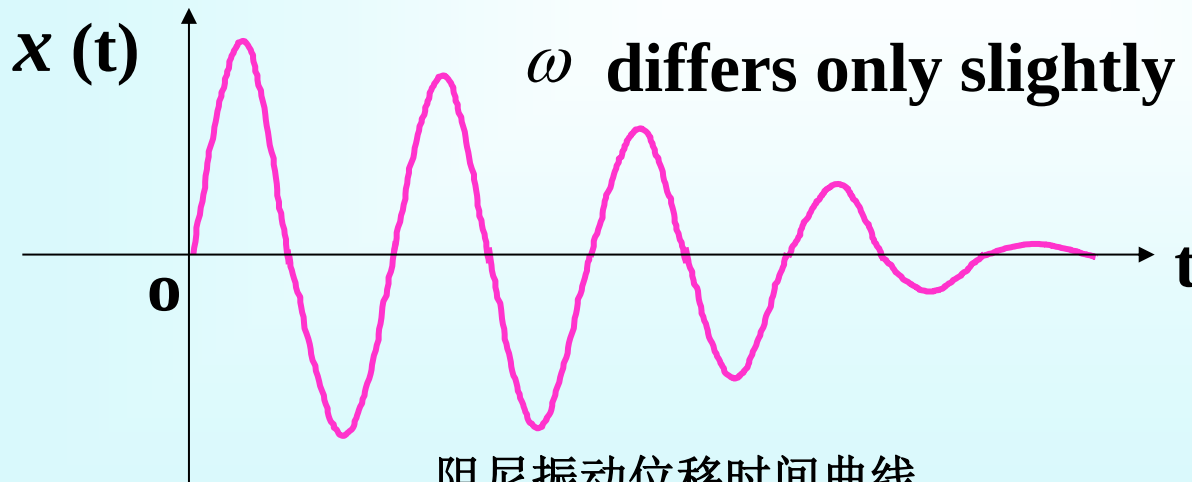
$$F_{\text{damping}} = -bv \quad ma = -kx - bv$$

$$m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = 0 \quad \frac{d^2 x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \omega_0^2 x = 0 \quad \omega_0 = \sqrt{\frac{k}{m}}$$

The damping constant b is small $x = Ae^{-\alpha t} \cos(\omega t + \phi)$

$$\alpha = b/2m \quad \omega = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} = \sqrt{\omega_0^2 - \frac{b^2}{4m^2}}$$

ω differs only slightly from ω_0 .



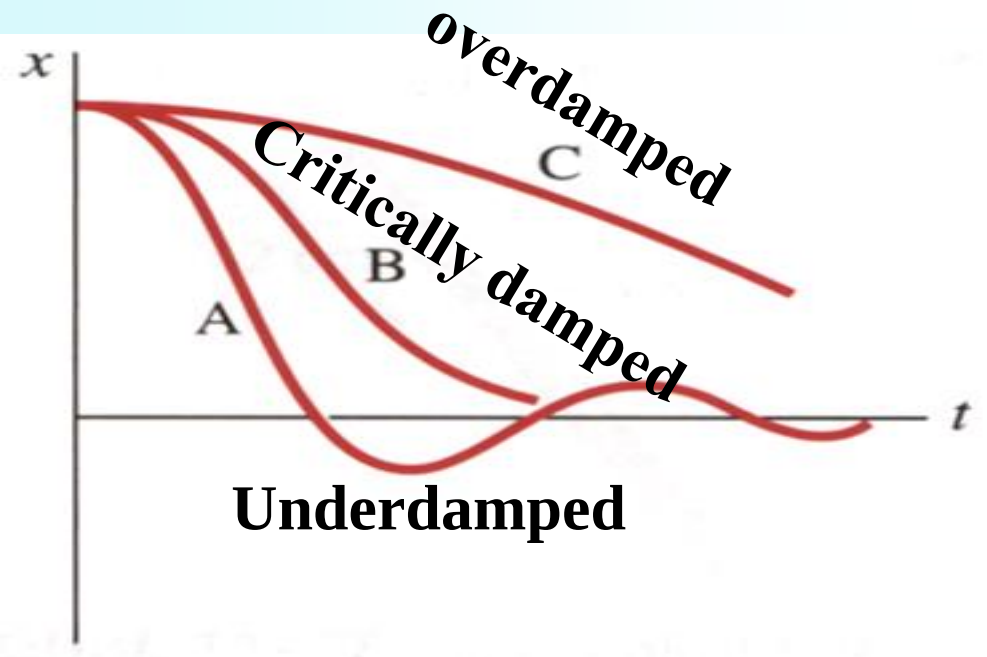
阻尼振动位移时间曲线

$$\text{if } b^2 < 4mk$$

In this case the system makes several swing before coming to rest, as curve **A**, called underdamped. (欠阻尼)

$$\text{if } b^2 > 4mk$$

In this case the system does not oscillate at all but returns directly to its equilibrium, as curve **C**, called overdamped. (过阻尼)

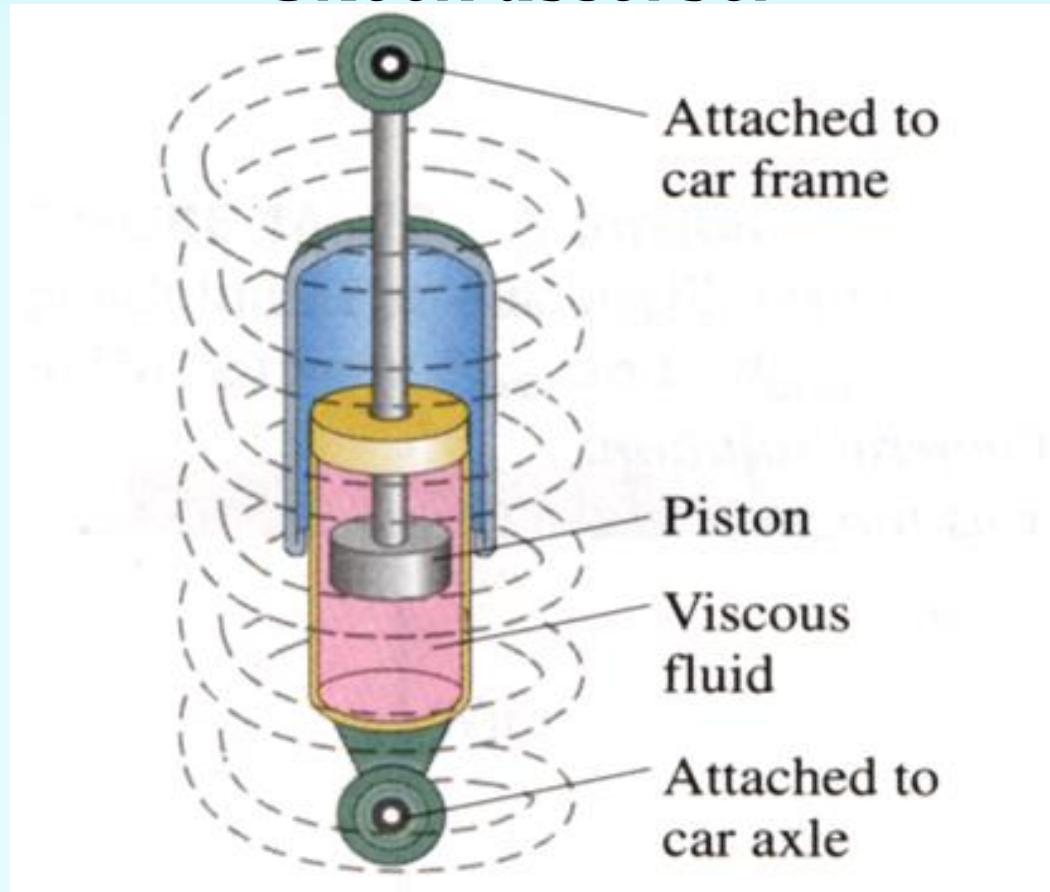


$$\text{if } b^2 = 4mk$$

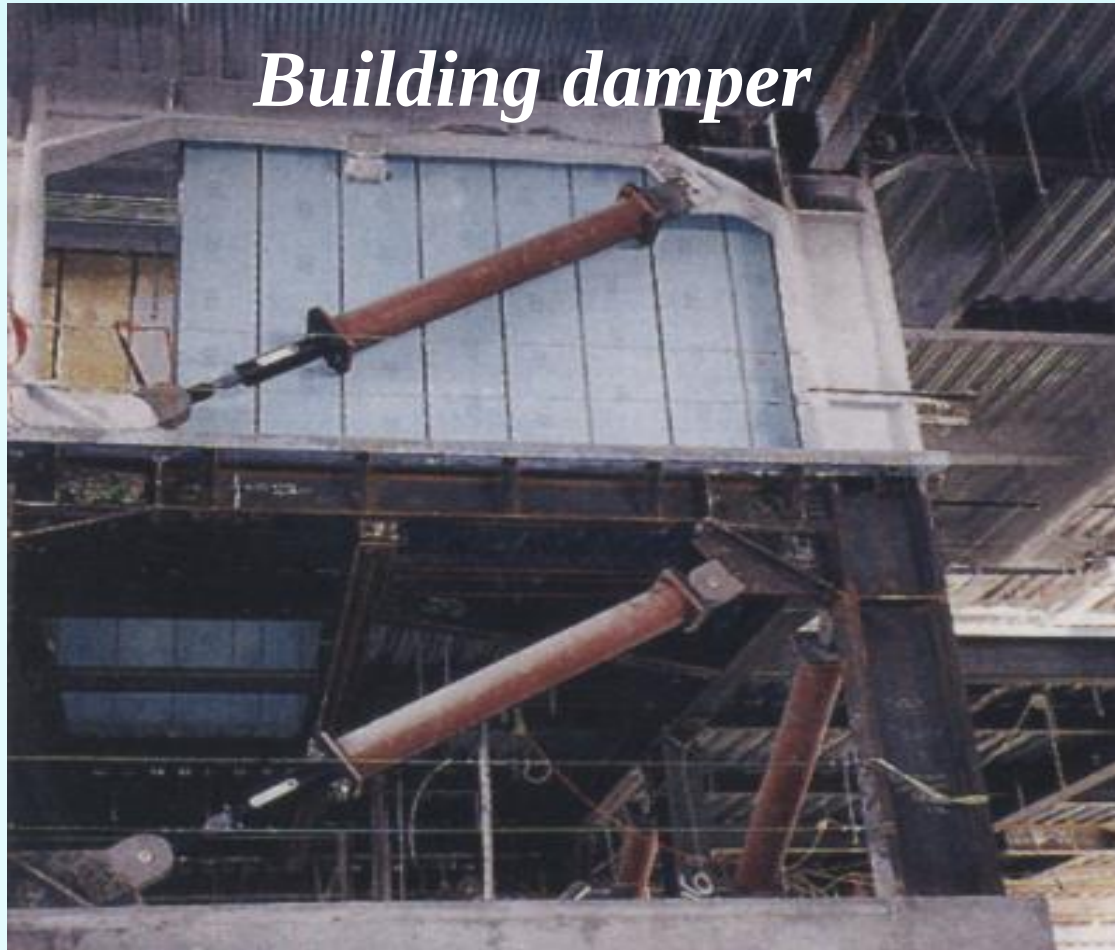
In this case equilibrium the system is reached in the shortest time, as curve **B**, called critical damping.

(临界阻尼)

Shock absorber



Automobile spring and shock absorber to provide damping so that car won't bounce up and down endlessly.



These huge dampers placed in a building structure look a lot like huge automobile shock absorbers, and they serve a similar purpose—to reduce the amplitude and the acceleration of moment when the shock of an earthquake hits.

***16.7 Forced Oscillations and Resonance**

Forced oscillations---

vibrations induced by external driving force.

Resonance ——— When a vibrating system is driven by a periodic force at the **natural frequency** of the system, amplitude of the vibrations becomes exceedingly large.

How can we make a child's swing getting higher and higher?

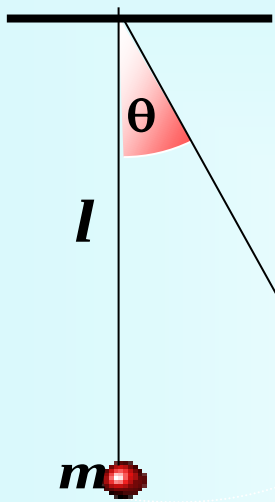
Forced oscillations of a swing by a force



In phase with the
nature frequency



Out of phase with
the nature frequency



If a force is applied, which oscillates with a frequency ω

$$F_{ext} = F_0 \cos \omega t$$

ω : Frequency of driving force

$$\frac{d^2 x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \omega_0^2 x = F_0 \cos \omega t$$

$$\omega_0 = \sqrt{k/m}$$

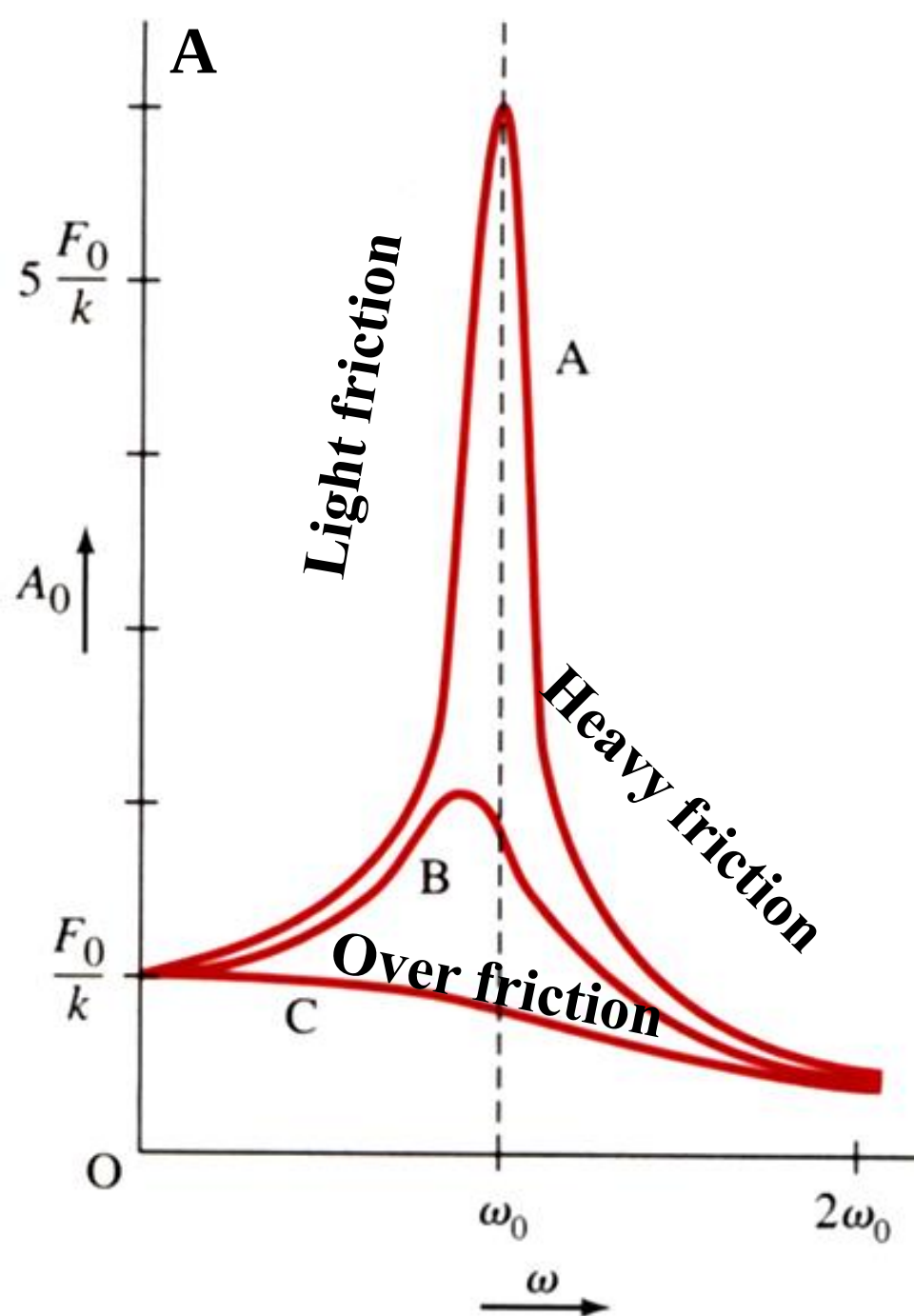
Natural frequency

$$x = A_0 \cos(\omega t + \phi_0)$$

$$A_0 = \frac{F_0}{m \sqrt{(\omega^2 - \omega_0^2)^2 + b^2 \omega^2 / m^2}}$$

$$\phi_0 = \tan^{-1} \frac{\omega_0^2 - \omega^2}{\omega(b/m)}$$

If we continue to push on oscillator with some frequency ω , it will eventually oscillate at that frequency.



$$A_0 = \frac{F_0}{m\sqrt{(\omega^2 - \omega_0^2)^2 + b^2\omega^2 / m^2}}$$

The larger the friction coefficient b becomes, the lower the peak height, as shown.

Amplitude of a forced harmonic oscillator as a function of ω .

ω/ω_0

$b=0$, (阻尼很小)

Let's interpret this equation for A:

$$A = \left| \frac{a_0}{\omega_0^2 - \omega^2} \right| \quad a_0 = \frac{F_0}{m}$$

(a) If $\omega \ll \omega_0$ $A \approx a_0 / \omega_0^2$

**This means the object will vibrate only slightly at frequency ω .
In other words, nothing spectacular happens.**

(b) If $\omega \gg \omega_0$ $A \approx a_0 / \omega^2$

The amplitude is even smaller. The object has no time to follow the driver.

(c) If $\omega \approx \omega_0$ $A \rightarrow \infty$

No matter how small a_0 is, as ω approaches ω_0 , resonance will happen. Energy is being continuously transferred into the system and none is dissipated.